Main Category	Category	Terms
Disease	OMIM_DISEASE	DNA topoisomerase II, resistance to inhibition of, by amsacrine
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Biological rhythms
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Isomerase
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Topoisomerase
Functional Annotations	UP_KW_PTM	Acetylation
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_KW_PTM	Isopeptide bond
Functional Annotations	UP_SEQ_FEATURE	ACT_SITE:O-(5'-phospho-DNA)-tyrosine intermediate
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic and acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cte
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cte
Functional Annotations	UP_SEQ_FEATURE	CROSSLNK:Glycyl lysine isopeptide (Lys-Gly) (interchain with G-Cte
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Histidine kinase/HSP90-like ATPase
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Topo IIA-type catalytic
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Toprim

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Nuclear export signal
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->A,C: Impairs bending of target DNA. Strongly reduce
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->A,C: Strongly reduced DNA cleavage.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:E->A,C: Impairs bending of target DNA. Strongly reduce
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:KKK->AAA: Reduced enzyme activity; abolishes stimula DNA binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:KKK->EEE: Strongly reduced enzyme activity; abolishes upon DNA binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->A: Abolishes binding to the antibody MPM2.
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with DNA
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with PLSCR1
Functional Annotations	UP_SEQ_FEATURE	SITE:Important for DNA bending; intercalates between base pairs
Functional Annotations	UP_SEQ_FEATURE	SITE:Interaction with DNA
Functional Annotations	UP_SEQ_FEATURE	SITE:Transition state stabilizer
Gene Ontology	GOTERM_BP_DIRECT	resolution of meiotic recombination intermediates
Gene Ontology	GOTERM_BP_DIRECT	sister chromatid segregation
Gene Ontology	GOTERM_BP_DIRECT	hematopoietic progenitor cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	DNA metabolic process
Gene Ontology	GOTERM_BP_DIRECT	DNA topological change
Gene Ontology	GOTERM_BP_DIRECT	chromatin organization
Gene Ontology	GOTERM_BP_DIRECT	DNA damage response
Gene Ontology	GOTERM_BP_DIRECT	chromosome segregation

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	female meiotic nuclear division
Gene Ontology	GOTERM_BP_DIRECT	chromosome condensation
Gene Ontology	GOTERM_BP_DIRECT	apoptotic chromosome condensation
Gene Ontology	GOTERM_BP_DIRECT	embryonic cleavage
Gene Ontology	GOTERM_BP_DIRECT	regulation of circadian rhythm
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of single stranded viral RNA replication via doul
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	rhythmic process
Gene Ontology	GOTERM_CC_DIRECT	nuclear chromosome
Gene Ontology	GOTERM_CC_DIRECT	chromosome, centromeric region
Gene Ontology	GOTERM_CC_DIRECT	condensed chromosome
Gene Ontology	GOTERM_CC_DIRECT	male germ cell nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	nucleolus
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	centriole
Gene Ontology	GOTERM_CC_DIRECT	DNA topoisomerase type II (double strand cut, ATP-hydrolyzing) co
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_CC_DIRECT	ribonucleoprotein complex

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	nucleotide binding
Gene Ontology	GOTERM_MF_DIRECT	magnesium ion binding
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	chromatin binding
Gene Ontology	GOTERM_MF_DIRECT	RNA binding
Gene Ontology	GOTERM_MF_DIRECT	DNA topoisomerase activity
Gene Ontology	GOTERM_MF_DIRECT	DNA topoisomerase type II (double strand cut, ATP-hydrolyzing) ac
Gene Ontology	GOTERM_MF_DIRECT	protein kinase C binding
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	ATP binding
Gene Ontology	GOTERM_MF_DIRECT	ATP-dependent activity, acting on DNA
Gene Ontology	GOTERM_MF_DIRECT	DNA binding, bending
Gene Ontology	GOTERM_MF_DIRECT	isomerase activity
Gene Ontology	GOTERM_MF_DIRECT	protein homodimerization activity
Gene Ontology	GOTERM_MF_DIRECT	ubiquitin binding
Gene Ontology	GOTERM_MF_DIRECT	metal ion binding
Gene Ontology	GOTERM_MF_DIRECT	protein heterodimerization activity
General Annotations	CHROMOSOME	17
General Annotations	CYTOGENETIC_LOCATION	17q21.2
General Annotations	ENTREZ_GENE_SUMMARY	This gene encodes a DNA topoisomerase, an enzyme that controls states of DNA during transcription. This nuclear enzyme is involve chromosome condensation, chromatid separation, and the relief o during DNA transcription and replication. It catalyzes the transient strands of duplex DNA which allows the strands to pass through o

Main Category	Category	Terms
		topology of DNA. Two forms of this enzyme exist as likely product The gene encoding this form, alpha, is localized to chromosome 1 localized to chromosome 3. The gene encoding this enzyme functi anticancer agents and a variety of mutations in this gene have bee development of drug resistance. Reduced activity of this enzyme r telangiectasia. [provided by RefSeq, Jul 2010]
General Annotations	UP_COMMENT_ACTIVITY_REGULATION	Specifically inhibited by the intercalating agent amsacrine.
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=4; Name=1; IsoId=P1 Name=2; IsoId=P11388-2; Sequence=VSP_006531; Name=3; Iso Sequence=VSP_006529; Name=4; IsoId=P11388-4; Sequence=VS
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=ATP-dependent breakage, passage and rejoining of doub EC=5.6.2.2; Evidence={ECO:0000255|PROSITE- ProRule:PRU0099 PubMed:19222228, ECO:0000269|PubMed:19697956};
General Annotations	UP_COMMENT_CATALYTIC_ACTIVITY	Reaction=ATP-dependent breakage, passage and rejoining of doub EC=5.6.2.2; Evidence={ECO:0000256|ARBA:ARBA00000185, ECO: RuleBase:RU362094};
General Annotations	UP_COMMENT_COFACTOR	Name=Ca(2+); Xref=ChEBI:CHEBI:29108; Evidence={ECO:000025 Name=Mg(2+); Xref=ChEBI:CHEBI:18420; Evidence={ECO:000025 Name=Mn(2+); Xref=ChEBI:CHEBI:29035; Evidence={ECO:000025
General Annotations	UP_COMMENT_COFACTOR	Name=Mg(2+); Xref=ChEBI:CHEBI:18420; Evidence={ECO:000025
General Annotations	UP_COMMENT_COFACTOR	Name=Mg(2+); Xref=ChEBI:CHEBI:18420; Evidence={ECO:00003 ECO:0000305|PubMed:19697956, ECO:0000305|PubMed:223236 PubMed:22841979}; Name=Mn(2+); Xref=ChEBI:CHEBI:29035; Ev PubMed:19222228, ECO:0000305|PubMed:19697956, ECO:00003 ECO:0000305|PubMed:22841979}; Name=Ca(2+); Xref=ChEBI:CHE Evidence={ECO:0000305|PubMed:19222228, ECO:0000305|PubMe PubMed:22323612, ECO:0000305|PubMed:22841979}; Note=Binds magnesium ions form salt bridges with both the protein and the DI divalent metal cations, such as Mn(2+) or Ca(2+). {ECO:0000305|P ECO:0000305|PubMed:19697956, ECO:0000305|PubMed:2232361 PubMed:22841979};
General Annotations	UP_COMMENT_DOMAIN	The N-terminus has several structural domains; the ATPase domai transducer domain (about 266-428) and the toprim domain (455-5 Comparing different structures shows ATP hydrolysis induces dom that are probably part of the mechanism of DNA cleavage and rejo {ECO:0000250|UniProtKB:P0AES6, ECO:0000269|PubMed:252029
General Annotations	UP_COMMENT_FUNCTION	Key decatenating enzyme that alters DNA topology by binding to tv molecules, generating a double-stranded break in one of the strand through the broken strand, and religating the broken strand (PubM PubMed:18790802, PubMed:22013166, PubMed:22323612). May period length of BMAL1 transcriptional oscillation (By similarity). { UniProtKB:Q01320, ECO:0000269|PubMed:17567603, ECO:00002 ECO:0000269|PubMed:22013166, ECO:0000269|PubMed:2232361
General Annotations	UP_COMMENT_FUNCTION	Key decatenating enzyme that alters DNA topology by binding to tv molecules, generating a double-stranded break in one of the strand through the broken strand, and religating the broken strand. May p period length of BMAL1 transcriptional oscillation. {ECO:0000256|
General Annotations	UP_COMMENT_INTERACTION	P11388; O14497-1: ARID1A; NbExp=2; IntAct=EBI-539628, EBI-159 BRCA1; NbExp=3; IntAct=EBI-539628, EBI-349905; P11388; P3522

Main Category	Category	Terms
		IntAct=EBI-539628, EBI-491549; P11388; Q05655: PRKCD; NbExp= EBI-704279;
General Annotations	UP_COMMENT_MISCELLANEOUS	Eukaryotic topoisomerase I and II can relax both negative and posi prokaryotic enzymes relax only negative supercoils.
General Annotations	UP_COMMENT_PTM	(Microbial infection) Deubiquitinated by Epstein-Barr virus BPLF1; SUMOylated TOP2A trapped in cleavage complexes, which halts th TOP2A-induced double-strand DNA breaks. {ECO:0000269|PubMe
General Annotations	UP_COMMENT_PTM	Phosphorylation has no effect on catalytic activity. However, phosp CSNK1D/CK1 promotes DNA cleavable complex formation. {ECO:0 ECO:0000269|PubMed:18062778, ECO:0000269|PubMed:1904307
General Annotations	UP_COMMENT_PTM	SUMOylated. {ECO:0000269|PubMed:34543352}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the type II topoisomerase family. {ECO:0000256|ARBA: ECO:0000256|RuleBase:RU362094}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the type II topoisomerase family. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cytoplasm {ECO:0000256|ARBA:ARBA00004496}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cytoplasm {ECO:0000269|PubMed:9155056}. Nucleus, nucleoplas PubMed:8299728}. Nucleus {ECO:0000269|PubMed:17567603, EC PubMed:22013166, ECO:0000269|PubMed:8299728, ECO:000026 nucleolus {ECO:0000269|PubMed:9155056}.
General Annotations	UP_COMMENT_SUBUNIT	Homodimer. Interacts with COPS5. Interacts with RECQL5; this stim Interacts with SETMAR; stimulates the topoisomerase activity (Pub PubMed:20457750). Interacts with DHX9; this interaction occurs in RNA- dependent manner, negatively regulates DHX9-mediated dou duplex helicase activity and stimulates TOP2A-mediated supercoil (PubMed:12711669). Interacts with HNRNPU (via C-terminus); this topoisomerase TOP2A from degradation and positively regulates t DNA in a RNA-dependent manner (By similarity). Interacts with MC (PubMed:23652018). Interacts with ERCC6 (PubMed:26030138). I (PubMed:17567603). Interacts with GCNA; this interaction allows t topoisomerase II (TOP2A) DNA-protein cross-links (By similarity). I (via dock II) and UBTF in the context of Pol I complex; may assist R releasing supercoils occurring during DNA unwinding. Interacts wi number of DNA damage response proteins, is recruited to sites of role in DNA damage repair (PubMed:23213405). {ECO:0000250|Un ECO:0000250|UniProtKB:Q01320, ECO:0000269|PubMed:1271166 PubMed:15126503, ECO:0000269|PubMed:16100112, ECO:00002 ECO:0000269|PubMed:20457750, ECO:0000269|PubMed:2201316 PubMed:22841979, ECO:0000269|PubMed:23213405, ECO:00002 ECO:0000269|PubMed:26030138, ECO:0000269|PubMed:3627149
General Annotations	UP_COMMENT_SUBUNIT	Homodimer. {ECO:0000256|RuleBase:RU362094}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed in the tonsil, spleen, lymph node, thymus, skin, pancreas brain and lung (PubMed:9155056). Also found in high-grade lymph tumors and seminomas (PubMed:9155056). {ECO:0000269|PubM
Identifiers	ENTREZ_GENE_ID	7153
Identifiers	ENSEMBL_GENE_ID	ENSG00000131747

Main Category	Category	Terms
Identifiers	UNIPROT_ID	TOP2A_HUMAN
Identifiers	UNIPROT_ID	J3QR57_HUMAN
Identifiers	UNIPROT_ID	J3KTB7_HUMAN
Identifiers	UNIPROT_ID	A0A4D6UXC9_HUMAN
Interactions	DRUGBANK	Moxifloxacin
Interactions	DRUGBANK	Amsacrine
Interactions	DRUGBANK	Dexrazoxane
Interactions	DRUGBANK	Valrubicin
Interactions	DRUGBANK	Teniposide
Interactions	DRUGBANK	Epirubicin
Interactions	DRUGBANK	Enoxacin
Interactions	DRUGBANK	Pefloxacin
Interactions	DRUGBANK	Ciprofloxacin
Interactions	DRUGBANK	Trovafloxacin
Interactions	DRUGBANK	Daunorubicin
Interactions	DRUGBANK	Etoposide
Interactions	DRUGBANK	Dactinomycin
Interactions	DRUGBANK	Lomefloxacin
Interactions	DRUGBANK	Doxorubicin
Interactions	DRUGBANK	Norfloxacin
Interactions	DRUGBANK	Ofloxacin
Interactions	DRUGBANK	Idarubicin
Interactions	DRUGBANK	Podofilox
Interactions	DRUGBANK	Mitoxantrone
Interactions	DRUGBANK	Sparfloxacin
Interactions	DRUGBANK	Genistein
Interactions	DRUGBANK	Fleroxacin
Interactions	DRUGBANK	Lucanthone
Interactions	DRUGBANK	Banoxantrone
Interactions	DRUGBANK	SP1049C
Interactions	DRUGBANK	Amonafide
Interactions	DRUGBANK	Elsamitrucin

Main Category	Category	Terms
Interactions	DRUGBANK	13-deoxydoxorubicin
Interactions	DRUGBANK	RTA 744
Interactions	DRUGBANK	Aldoxorubicin
Interactions	DRUGBANK	ZEN-012
Interactions	DRUGBANK	Amrubicin
Interactions	DRUGBANK	Becatecarin
Interactions	DRUGBANK	Annamycin
Interactions	DRUGBANK	Declopramide
Interactions	DRUGBANK	Finafloxacin
Interactions	PUBCHEM	2-[4-[5-[4-(1,4,5,6-Tetrahydropyrimidin-2-yl)phenyl]furan-2-yl]phenyl] tetrahydropyrimidine;dihydrochloride
Interactions	PUBCHEM	N-[3-(acridin-9-ylamino)-5-(hydroxymethyl)phenyl]-4-(dimethylamin
Interactions	PUBCHEM	[3-(acridin-9-ylamino)-5-aminophenyl]methyl N-methylcarbamate
Interactions	PUBCHEM	14-[(3S)-3-aminopyrrolidin-1-yl]-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.05,10.017,21]henicosa-2(11),3,5,7,9,13 carboxylic acid
Interactions	PUBCHEM	N-[3-(acridin-9-ylamino)-5-(pyridin-1-ium-1-ylmethyl)phenyl]methan
Interactions	PUBCHEM	N-[(7S)-1,2,3-trihydroxy-10-(methylamino)-9-oxo-6,7-dihydro-5H-ben
Interactions	PUBCHEM	tert-butyl N-[3-(acridin-9-ylamino)-5-(methylcarbamoyloxymethyl)p
Interactions	PUBCHEM	[3-(Acridin-9-ylamino)-5-(methanesulfonamido)phenyl]methyl acet
Interactions	PUBCHEM	N,O-bis(acetonylacetyl)-3-(9-acridinylamino)-5-hydroxymethylanilin
Interactions	PUBCHEM	14-(3-Aminopyrrolidin-1-yl)-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.05,10.017,21]henicosa-2(11),3,5,7,9,13 carboxylic acid
Interactions	PUBCHEM	N-[3-(acridin-9-ylamino)-5-(hydroxymethyl)phenyl]methanesulfonar
Interactions	PUBCHEM	N-[(7S)-7-acetamido-1,2,3-trihydroxy-9-oxo-6,7-dihydro-5H-benzo[a] chlorobenzamide
Interactions	PUBCHEM	Cystodytin J
Interactions	PUBCHEM	N-[[14-[2-(diethylamino)ethyl]-4-hydroxy-8-thia-14,15-diazatetracyclo[7.6.1.02,7.013,16]hexadeca-1(15),2(7),3,5,9,11,13(yl]methyl]methanesulfonamide
Interactions	PUBCHEM	10-[(3S)-3-Aminopyrrolidin-1-yl]-11-fluoro-14-oxo-8-oxa-1-azatetracyclo[7.7.1.02,7.013,17]heptadeca-2,4,6,9(17),10,12,15-hep
Interactions	PUBCHEM	methyl N-[[1-[2-(diethylamino)ethylamino]-7-hydroxy-9-oxothioxant
Interactions	PUBCHEM	methyl N-[[14-[2-(diethylamino)ethyl]-4-hydroxy-8-thia-14,15-diazatetracyclo[7.6.1.02,7.013,16]hexadeca-1(15),2(7),3,5,9,11,13(yl]methyl]carbamate

Main Category	Category	Terms
Interactions	PUBCHEM	N-[[1-[2-(diethylamino)ethylamino]-7-hydroxy-9-oxothioxanthen-4-yl
Interactions	PUBCHEM	14-(3-Aminopyrrolidin-1-yl)-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.03,8.017,21]henicosa-2(11),3,5,7,9,13(carboxylic acid
Interactions	PUBCHEM	14-(3-Aminopyrrolidin-1-yl)-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.04,9.017,21]henicosa-2,4,6,8,10,13(21) carboxylic acid
Interactions	PUBCHEM	15-Fluoro-18-oxo-14-piperazin-1-yl-12-oxa-1-azapentacyclo[11.7.1.02,11.03,8.017,21]henicosa-2(11),3,5,7,9,13(carboxylic acid
Interactions	PUBCHEM	5-[(5R,5aR,8aR,9R)-5-hydroxy-8-oxo-5a,6,8a,9-tetrahydro-5H-[2]benz yl]-3-methoxycyclohexa-3,5-diene-1,2-dione
Interactions	PUBCHEM	Ethyl curcumin
Interactions	PUBCHEM	1-[2-(Diethylamino)ethylamino]-7-hydroxy-4-(methylaminomethyl)th
Interactions	PUBCHEM	15-Fluoro-18-oxo-14-piperazin-1-yl-12-oxa-1-azapentacyclo[11.7.1.02,11.05,10.017,21]henicosa-2(11),3,5,7,9,13 carboxylic acid
Interactions	PUBCHEM	15-Fluoro-18-oxo-14-piperazin-1-yl-12-oxa-1-azapentacyclo[11.7.1.02,11.04,9.017,21]henicosa-2,4,6,8,10,13(21) carboxylic acid
Interactions	PUBCHEM	3Alpha,27-Dihydroxylup-20(29)-En-28-Oic Acid Methyl Ester
Interactions	PUBCHEM	CP 115953
Interactions	PUBCHEM	Dehydroeburiconic Acid
Interactions	PUBCHEM	9-Chloro-6-piperazin-1-ylthiochromeno[2,3-c]quinolin-12-one
Interactions	PUBCHEM	9-(3,4-Dimethoxy-phenyl)-8-H-furo(3',4':6,7)naphtho(2,3-d)(1,3)diox
Interactions	PUBCHEM	NPF-etoposide
Interactions	PUBCHEM	6-amino-7-(1H-benzimidazol-2-yl)-5-[4-(dimethylamino)butyl]pyrrol dicarbonitrile
Interactions	PUBCHEM	2,6,11-Trimethylpyrido[4,3-b]carbazol-2-ium iodide
Interactions	PUBCHEM	3,6-Diamino-9-(4-(methylsulfonyl)aminophenyl)aminoacridine
Interactions	PUBCHEM	Win 58161
Interactions	PUBCHEM	1-Cyclopropyl-6,8-difluoro-1,4-dihydro-7-(2,6-dimethyl-4-pyridinyl)-4
Interactions	PUBCHEM	N-[(5S,5aS,8aR,9R)-8-oxo-9-(3,4,5-trimethoxyphenyl)-5a,6,8a,9-tetra [1,3]benzodioxol-5-yl]-1-(4-chlorophenyl)triazole-4-carboxamide
Interactions	PUBCHEM	1-[(2R,4S,5S)-5-[[tert-butyl(dimethyl)silyl]oxymethyl]-4-[2-(3,4-dimet yl]oxolan-2-yl]-5-methylpyrimidine-2,4-dione
Interactions	PUBCHEM	1-[(2R,4S,5S)-5-[[tert-butyl(dimethyl)silyl]oxymethyl]-4-[(3,4-dioxocy yl)methyl]oxolan-2-yl]-5-methylpyrimidine-2,4-dione
Interactions	PUBCHEM	N-[4-(dimethylamino)butyl]-11-oxoindeno[1,2-c]cinnoline-2-carboxa

Main Category	Category	Terms
Interactions	PUBCHEM	16-[2-(Dimethylamino)ethoxy]-11,12-diazapentacyclo[11.8.0.02,10.04,9.014,19]henicosa-1,4,6,8,10,12,1
Interactions	PUBCHEM	4-Methylamino-[1,2,4]triazolo[4,3-a]quinoxaline
Interactions	PUBCHEM	N',N'-dimethyl-N-quinolino[4,3-b]quinoxalin-6-ylbutane-1,4-diamine
Interactions	PUBCHEM	5-Amino-11-fluoro-14-oxo-10-(4-piperidin-1-ylpiperidin-1-yl)-8-oxa-1 azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	10-[(3S)-3-aminopyrrolidin-1-yl]-11-fluoro-5-nitro-14-oxo-8-oxa-1-azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	5-Amino-11-fluoro-10-(2-morpholin-4-ylethylamino)-14-oxo-8-oxa-1 azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	5-amino-10-[(3R)-3-aminopyrrolidin-1-yl]-11-fluoro-14-oxo-8-oxa-1-azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	5-amino-10-[(3S)-3-aminopyrrolidin-1-yl]-11-fluoro-14-oxo-8-oxa-1-azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	2,4,5,8,9,11-Hexazatetracyclo[10.4.0.02,6.07,11]hexadeca-1(16),5,7
Interactions	PUBCHEM	N4-(4-(dimethylamino)phenyl)-2-(pyridin-4-yl)-5,6,7,8-tetrahydroqui
Interactions	PUBCHEM	Hymenialdisine
Interactions	PUBCHEM	(NZ)-N-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquino
Interactions	PUBCHEM	2-[[1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-4-
Interactions	PUBCHEM	5-(3-Fluoro-4,5-dihydroxy-6-methyloxan-2-yl)oxy-6-hydroxynaphtho
Interactions	PUBCHEM	4-Hydroxy-3-[(4-hydroxy-2-oxochromen-3-yl)-(4-nitrophenyl)methyl]
Interactions	PUBCHEM	3-[(4-Chlorophenyl)-(4-hydroxy-2-oxochromen-3-yl)methyl]-4-hydrox
Interactions	PUBCHEM	3-[3-(3-Aminopropylamino)-2-hydroxypropoxy]-1-hydroxyxanthen-9
Interactions	PUBCHEM	6-[4-(3-Aminopropylamino)butyl]-13-[3-[(2-methoxyphenyl)methyla diazatetracyclo[6.6.2.04,16.011,15]hexadeca-1(15),2,4(16),8,10-pe
Interactions	PUBCHEM	3-[3-[3-[4-(3-Aminopropylamino)butylamino]propylamino]-2-hydrox one;tetrahydrochloride
Interactions	PUBCHEM	3-[3-[3-(4-Aminobutylamino)propylamino]-2-hydroxypropoxy]-1-hyd trifluoroacetic acid)
Interactions	PUBCHEM	3-[1,3-Benzodioxol-5-yl-(4-hydroxy-2-oxochromen-3-yl)methyl]-4-hy
Interactions	PUBCHEM	N-[3-(dimethylamino)propyl]-11-oxoindeno[1,2-c]cinnoline-2-carbox
Interactions	PUBCHEM	3-[(2-Fluorophenyl)-(4-hydroxy-2-oxochromen-3-yl)methyl]-4-hydrox
Interactions	PUBCHEM	6-[3-(3-Aminopropylamino)propyl]-13-[3-[(2-methoxyphenyl)methyl diazatetracyclo[6.6.2.04,16.011,15]hexadeca-1(15),2,4(16),8,10-pe
Interactions	PUBCHEM	3-[(4-Bromophenyl)-(4-hydroxy-2-oxochromen-3-yl)methyl]-4-hydrox
Interactions	PUBCHEM	6-[3-[4-(3-Aminopropoxy)butoxy]propyl]-13-[3-[(2-methoxyphenyl)m diazatetracyclo[6.6.2.04,16.011,15]hexadeca-1(15),2,4(16),8,10-pe

Main Category	Category	Terms
Interactions	PUBCHEM	4-hydroxy-6-oxo-N,1,3-triphenyl-2-sulfanylidenepyrimidine-5-carbox
Interactions	PUBCHEM	4-Hydroxy-3-[(4-hydroxy-2-oxochromen-3-yl)-phenylmethyl]naphtha
Interactions	PUBCHEM	6-[3-(4-Aminobutylamino)propyl]-13-[3-[(2-methoxyphenyl)methyla diazatetracyclo[6.6.2.04,16.011,15]hexadeca-1(15),2,4(16),8,10-pe
Interactions	PUBCHEM	3-[(4-Fluorophenyl)-(4-hydroxy-2-oxochromen-3-yl)methyl]-4-hydrox
Interactions	PUBCHEM	6-[3-[4-(3-Aminopropylamino)butylamino]propyl]-13-[3-[(2-methoxyphenyl)methylamino]propyl]-6,13-diazatetracyclo[6.6.2.04,16.011,15]hexadeca-1(15),2,4(16),8,10-pe
Interactions	PUBCHEM	3-[3-(4-Aminobutylamino)-2-hydroxypropoxy]-1-hydroxyxanthen-9-c acid)
Interactions	PUBCHEM	N-[2-(12-methoxy-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca octaen-10-yl)ethyl]-3-methylbut-2-enamide
Interactions	PUBCHEM	5-[(5S,5aR,8aR,9R)-5-hydroxy-8-oxo-5a,6,8a,9-tetrahydro-5H-[2]benz yl]-3-methoxycyclohexa-3,5-diene-1,2-dione
Interactions	PUBCHEM	5-Amino-11-fluoro-10-(4-methylpiperazin-1-yl)-14-oxo-8-oxa-1-azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	N-[2-(11-methylsulfanyl-12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca-1(17),2,4,6,8,10,13,15 yl)ethyl]propanamide
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-5-(3-hydroxyanilino)-9-(4-hydroxy-3,5-dimethoxyph [2]benzofuro[5,6-f][1,3]benzodioxol-8-one
Interactions	PUBCHEM	(2E,4E)-7-[4-[2-[(2S,3S)-2-[[tert-butyl(dimethyl)silyl]oxymethyl]-5-oxo yl]phenyl]-N-hydroxy-4,6-dimethyl-7-oxohepta-2,4-dienamide
Interactions	PUBCHEM	2-[[4-[(3,4-dichloro-5-methyl-1H-pyrrole-2-carbonyl)amino]-3-propan acid
Interactions	PUBCHEM	(2S)-2-[[4-[(3,4-dichloro-5-methyl-1H-pyrrole-2-carbonyl)amino]-3-pr phenylpropanoic acid
Interactions	PUBCHEM	N-butyl-[1,2,4]triazolo[4,3-a]quinoxalin-4-amine
Interactions	PUBCHEM	1-Phenyl-3-([1,2,4]triazolo[4,3-a]quinoxalin-4-ylamino)urea
Interactions	PUBCHEM	1-Phenyl-3-([1,2,4]triazolo[4,3-a]quinoxalin-4-ylamino)thiourea
Interactions	PUBCHEM	N-[(E)-(2,6-dichlorophenyl)methylideneamino]-[1,2,4]triazolo[4,3-a]c
Interactions	PUBCHEM	1-Cyclohexyl-3-([1,2,4]triazolo[4,3-a]quinoxalin-4-ylamino)urea
Interactions	PUBCHEM	1-Ethyl-3-([1,2,4]triazolo[4,3-a]quinoxalin-4-ylamino)thiourea
Interactions	PUBCHEM	2-(2,4,5,8,9,11-hexazatetracyclo[10.4.0.02,6.07,11]hexadeca-1(16), ylsulfanyl)-N-(3-methoxyphenyl)acetamide
Interactions	PUBCHEM	N-(4-fluorophenyl)-2-(2,4,5,8,9,11-hexazatetracyclo[10.4.0.02,6.07,11]hexadeca-1(16),3,5,7,9,12,14-h
Interactions	PUBCHEM	Ethyl 2-(2,4,5,8,9,11-hexazatetracyclo[10.4.0.02,6.07,11]hexadeca-1 ylsulfanyl)acetate
Interactions	PUBCHEM	4-[[(5S,5aS,8aR,9R)-9-(4-hydroxy-3,5-dimethoxyphenyl)-8-oxo-5a,6,8

Main Category	Category	Terms
		[2]benzofuro[5,6-f][1,3]benzodioxol-5-yl]amino]benzonitrile
Interactions	PUBCHEM	ethyl 4-[[(5S,5aS,8aR,9R)-9-(4-hydroxy-3,5-dimethoxyphenyl)-8-oxo- [2]benzofuro[5,6-f][1,3]benzodioxol-5-yl]amino]benzoate
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-5-(1,3-benzodioxol-5-ylamino)-9-(4-hydroxy-3,5-din tetrahydro-5H-[2]benzofuro[5,6-f][1,3]benzodioxol-8-one
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-5-(2,3-dihydro-1,4-benzodioxin-6-ylamino)-9-(4-hyc dimethoxyphenyl)-5a,6,8a,9-tetrahydro-5H-[2]benzofuro[5,6-f][1,3]b
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-9-(4-hydroxy-3,5-dimethoxyphenyl)-5-(4-morpholin tetrahydro-5H-[2]benzofuro[5,6-f][1,3]benzodioxol-8-one
Interactions	PUBCHEM	GL-331
Interactions	PUBCHEM	9-((4-Aminophenyl)amino)-5,8,8a,9-tetrahydro-5-(4-hydroxy-3,5-dim furo(3',4':6,7)naphtho(2,3-d)-1,3-dioxol-6(5aH)-one monohydrochlo (5alpha,5abeta,8aalpha,9beta))-
Interactions	PUBCHEM	5-Amino-1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-4-ox
Interactions	PUBCHEM	5-Amino-11-fluoro-14-oxo-10-piperazin-1-yl-8-oxa-1- azatetracyclo[7.7.1.02,7.013,17]heptadeca-2(7),3,5,9(17),10,12,15-
Interactions	PUBCHEM	2-[[3,4-bis(chloranyl)-5-methyl-1
Interactions	PUBCHEM	(E)-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-4
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-(phenylmet
Interactions	PUBCHEM	3-[(4-Chlorophenyl)methyl]-1-cyclopropyl-7-(2,6-dimethyl-4-pyridiny
Interactions	PUBCHEM	N-(8-hydroxy-4-phenylquinolin-7-yl)-2,2-dimethylpropanamide
Interactions	PUBCHEM	N-[4-(1-benzofuran-2-yl)-8-hydroxyquinolin-7-yl]acetamide
Interactions	PUBCHEM	N-[4-(5-fluoro-2-hydroxyphenyl)-8-hydroxyquinolin-7-yl]acetamide
Interactions	PUBCHEM	6-N-tert-butyl-8-ethyl-2-N-[2-(2-morpholin-4-ylethoxy)quinolin-6-yl]-7
Interactions	PUBCHEM	N-[4-(5-fluoro-2-hydroxyphenyl)-8-hydroxyquinolin-7-yl]-3-morpholin
Interactions	PUBCHEM	N-[8-hydroxy-4-[2-hydroxy-5-(trifluoromethyl)phenyl]quinolin-7-yl]ac
Interactions	PUBCHEM	N-[4-(furan-3-yl)-8-hydroxyquinolin-7-yl]acetamide
Interactions	PUBCHEM	N-(8-hydroxy-4-thiophen-3-ylquinolin-7-yl)acetamide
Interactions	PUBCHEM	N-(8-hydroxy-4-thiophen-2-ylquinolin-7-yl)acetamide
Interactions	PUBCHEM	N-[8-hydroxy-4-[3-(trifluoromethyl)phenyl]quinolin-7-yl]acetamide
Interactions	PUBCHEM	4-[(4-chlorophenyl)methylsulfanyl]-6-[(4-propan-2-ylphenyl)methyls
Interactions	PUBCHEM	N-[3-(1,1-difluoroethyl)phenyl]-4-hydroxy-6-oxo-1,3-diphenyl-2-sulfa carboxamide
Interactions	PUBCHEM	N-[4-(1-benzofuran-2-yl)-8-hydroxyquinolin-7-yl]-2-morpholin-4-ylac
Interactions	PUBCHEM	(2-Chloroquinolin-3-yl)-thiophen-2-ylmethanol
Interactions	PUBCHEM	4-hydroxy-N-[(E)-3-(3-methoxyphenyl)prop-2-enyl]-6-oxo-1,3-diphen

Main Category	Category	Terms
		carboxamide
Interactions	PUBCHEM	N-[2-(7-acetamido-8-hydroxy-5-methoxyquinolin-4-yl)phenyl]benzar
Interactions	PUBCHEM	N-[4-(7-acetamido-8-hydroxyquinolin-4-yl)phenyl]acetamide
Interactions	PUBCHEM	6-[(2-fluorophenyl)methylsulfanyl]-4-[[3-fluoro-5-(trifluoromethyl)ph triazin-2-one
Interactions	PUBCHEM	N-[4-(3-fluorophenyl)-8-hydroxyquinolin-7-yl]acetamide
Interactions	PUBCHEM	2-Chloro-3-[(4-fluorophenyl)methoxy-thiophen-2-ylmethyl]quinoline
Interactions	PUBCHEM	2-Chloro-3-[(4-chlorophenyl)methoxy-thiophen-2-ylmethyl]quinoline
Interactions	PUBCHEM	N-[4-(1-benzofuran-2-yl)-8-hydroxyquinolin-7-yl]-3-morpholin-4-ylpro
Interactions	PUBCHEM	6-Hydroxy-4-oxo-1,3-diphenyl-2-thioxo-N-(3-(trifluoromethoxy)phen tetrahydropyrimidine-5-carboxamide
Interactions	PUBCHEM	N-[8-hydroxy-4-[4-(trifluoromethyl)phenyl]quinolin-7-yl]acetamide
Interactions	PUBCHEM	4-hydroxy-6-oxo-1,3-diphenyl-N-(3-propan-2-yloxyphenyl)-2-sulfanyl carboxamide
Interactions	PUBCHEM	6-[(2-fluorophenyl)methylsulfanyl]-4-[(4-propan-2-ylphenyl)methyls
Interactions	PUBCHEM	6-[(4-chlorophenyl)methylsulfanyl]-4-[[3-fluoro-5-(trifluoromethyl)ph triazin-2-one
Interactions	PUBCHEM	N-(4-oxo-2-phenylchromen-7-yl)-2,5-bis(trifluoromethyl)benzenesul
Interactions	PUBCHEM	2,4,6-trimethyl-N-(4-oxo-2-phenylchromen-7-yl)benzenesulfonamid
Interactions	PUBCHEM	3,4-dimethoxy-N-(4-oxo-2-phenylchromen-7-yl)benzenesulfonamid
Interactions	PUBCHEM	[3-(Acridin-9-ylamino)phenyl]methanol
Interactions	PUBCHEM	Topoisomerase II inhibitor 9
Interactions	PUBCHEM	Topoisomerase II|A-IN-4
Interactions	PUBCHEM	N-(4-carbamoyl-2-propan-2-yloxyphenyl)-3,4-dichloro-5-methyl-1H-
Interactions	PUBCHEM	4-[(2R,3S)-3-(3-oxopiperazin-1-yl)butan-2-yl]piperazine-2,6-dione
Interactions	PUBCHEM	4-[8-Amino-4-[2-(trifluoromethyl)phenyl]-5,6-dihydrobenzo[h]quinoli
Interactions	PUBCHEM	Topoisomerase II|A-IN-7
Interactions	PUBCHEM	12-Butan-2-yl-11-imino-9-(4-methoxyphenyl)-2-oxa-4,6,12,14-tetrazatricyclo[8.4.0.03,8]tetradeca-1(10),3(8),13-triene-5,7-dione
Interactions	PUBCHEM	N-[4-(1,3-thiazol-2-ylsulfamoyl)phenyl]-2-([1,2,4]triazolo[4,3-a]quino
Interactions	PUBCHEM	(3E)-1-cyclopropyl-3-[[1-cyclopropyl-6-fluoro-7-(4-methylpiperazin-1 dihydroquinolin-3-yl]methylidene]-6-fluoro-7-piperazin-1-yl-2H-quino
Interactions	PUBCHEM	7-imino-9-(4-methoxyphenyl)-11,13-dioxo-N-phenyl-2-oxa-4,6,12,14 tetrazatricyclo[8.4.0.03,8]tetradeca-1(10),3(8),4-triene-6-carbothioa
Interactions	PUBCHEM	12-(2-Hydroxyphenyl)-11-imino-9-(4-methoxyphenyl)-2-oxa-4,6,12,1 tetrazatricyclo[8.4.0.03,8]tetradeca-1(10),3(8),13-triene-5,7-dione

Main Category	Category	Terms
Interactions	PUBCHEM	N-(4-carbamoyl-2-propan-2-yloxyphenyl)-4-chloro-5-ethyl-1H-imida
Interactions	PUBCHEM	3,4-dichloro-5-methyl-N-[4-(piperidin-4-ylcarbamoyl)-2-propan-2-ylo carboxamide;hydrochloride
Interactions	PUBCHEM	3-[[3-[[5-carbamoyl-2-[(3,4-dichloro-5-methyl-1H-pyrrole-2-carbonyl)amino]phenoxy]methyl]benzoyl]amino]propanoic acid
Interactions	PUBCHEM	N-[2-[2-(3-aminopiperidin-1-yl)-2-oxoethoxy]-4-carbamoylphenyl]-3, pyrrole-2-carboxamide;2,2,2-trifluoroacetic acid
Interactions	PUBCHEM	N-[2-[[4-(2-aminoethylcarbamoyl)phenyl]methoxy]-4-carbamoylphe pyrrole-2-carboxamide;dihydrochloride
Interactions	PUBCHEM	N-(5-carbamoyl-2-piperidin-4-yloxyphenyl)-3,4-dichloro-5-methyl-1H carboxamide;hydrochloride
Interactions	PUBCHEM	(E)-N-[4-(6-hydroxynaphthalen-2-yl)phenyl]-3-phenylprop-2-enamide
Interactions	PUBCHEM	3,4-dichloro-5-methyl-N-[4-(piperidin-4-ylmethylcarbamoyl)-2-propa carboxamide;hydrochloride
Interactions	PUBCHEM	N-[2-[[3-(2-aminoethylcarbamoyl)phenyl]methoxy]-4-carbamoylphe pyrrole-2-carboxamide;hydrochloride
Interactions	PUBCHEM	3,4-dichloro-5-methyl-N-[2-[[3-(pyrrolidin-3-ylcarbamoyl)phenyl]met carboxamide;hydrochloride
Interactions	PUBCHEM	N-(4-carbamoyl-2-piperidin-4-yloxyphenyl)-3,4-dichloro-5-methyl-1H carboxamide;hydrochloride
Interactions	PUBCHEM	3,4-dichloro-5-methyl-N-[4-(piperidin-3-ylcarbamoyl)-2-propan-2-ylo carboxamide;hydrochloride
Interactions	PUBCHEM	3,4-dichloro-5-methyl-N-[2-[[3-(piperidin-3-ylcarbamoyl)phenyl]meth carboxamide;hydrochloride
Interactions	PUBCHEM	2-[[3-[[5-carbamoyl-2-[(3,4-dichloro-5-methyl-1H-pyrrole-2-carbonyl)amino]phenoxy]methyl]benzoyl]amino]acetic acid
Interactions	PUBCHEM	N-[4-(2-aminoethylcarbamoyl)-2-propan-2-yloxyphenyl]-3,4-dichloro carboxamide;hydrochloride
Interactions	PUBCHEM	Diplamine
Interactions	PUBCHEM	2-(4-Amino-cyclohexyl)-5-cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6 pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	Ascididemin
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-[(4-hydroxy
Interactions	PUBCHEM	2-[4-[5-[4-(3a,4,5,6,7,7a-hexahydro-1H-benzimidazol-2-yl)phenyl]fura hexahydro-1H-benzimidazole;dihydrochloride
Interactions	PUBCHEM	Amsacrine
Interactions	PUBCHEM	Aconitin
Interactions	PUBCHEM	N,N'-bis[2-[2-(6-chloro-8-oxo-14,15-diazatetracyclo[7.6.1.02,7.013,16]hexadeca-1(15),2(7),3,5,9(16),10 methylamino]ethyl]butanediamide

Main Category	Category	Terms
Interactions	PUBCHEM	F-14512
Interactions	PUBCHEM	Bns-22
Interactions	PUBCHEM	Ciprofloxacin
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-5-[4-(1H-benzimidazol-2-yl)anilino]-9-(4-hydroxy-3, tetrahydro-5H-isobenzofuro[5,6-f][1,3]benzodioxol-8-one
Interactions	PUBCHEM	N-[4-[[(5S,5aS,8aR,9R)-9-(4-hydroxy-3,5-dimethoxy-phenyl)-8-oxo-5a isobenzofuro[5,6-f][1,3]benzodioxol-5-yl]amino]phenyl]-1,7,7-trimeth oxabicyclo[2.2.1]heptane-4-carboxamide
Interactions	PUBCHEM	Daunorubicin
Interactions	PUBCHEM	Win 33377
Interactions	PUBCHEM	Doxorubicin
Interactions	PUBCHEM	Ellipticine
Interactions	PUBCHEM	Etoposide
Interactions	PUBCHEM	5-Cyclopropyl-2-(4-dimethylamino-cyclohexyl)-7-(2,6-dimethyl-pyric dihydro-pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	3-(9-Acridinylamino)-5-hydroxymethylaniline
Interactions	PUBCHEM	N-[(7S)-1,2,3-trihydroxy-10-methylsulfanyl-9-oxo-6,7-dihydro-5H-ben
Interactions	PUBCHEM	18-[(3S)-3-aminopyrrolidin-1-yl]-19-fluoro-22-oxo-16-oxa-1-azahexacyclo[15.7.1.02,15.05,14.06,11.021,25]pentacosa-2(15),3,5 undecaene-23-carboxylic acid
Interactions	PUBCHEM	(R)-13-(3-aminopyrrolidin-1-yl)-12-fluoro-10-oxo-10H-naphtho[1,2-c carboxylic acid
Interactions	PUBCHEM	Epirubicin
Interactions	PUBCHEM	Mitoxantrone
Interactions	PUBCHEM	Idarubicin
Interactions	PUBCHEM	N2-(benzo[d]thiazol-6-yl)-N6-tert-butyl-9H-purine-2,6-diamine
Interactions	PUBCHEM	N2-(benzo[d]thiazol-6-yl)-N6-cycloheptyl-9H-purine-2,6-diamine
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-(pyridin-4-y
Interactions	PUBCHEM	N-[2-[[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-4-oxoqu yl]methyl]phenyl]-2,2,2-trifluoroacetamide
Interactions	PUBCHEM	1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-(1H-pyrrol-2
Interactions	PUBCHEM	N-[2-[[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-4-oxoqu yl]methyl]phenyl]acetamide
Interactions	PUBCHEM	3-[(2-Aminophenyl)methyl]-1-cyclopropyl-7-(2,6-dimethyl-4-pyridiny
Interactions	PUBCHEM	3-[(4-Aminophenyl)methyl]-1-cyclopropyl-7-(2,6-dimethyl-4-pyridiny
Interactions	PUBCHEM	1-Cyclopropyl-3-[(2,3-dihydroxyphenyl)methyl]-7-(2,6-dimethyl-4-pyr one

Main Category	Category	Terms
Interactions	PUBCHEM	N-[2-[[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-4-oxoqu yl]methyl]phenyl]methanesulfonamide
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-[[2-(hydroxymethyl)phenyl]methyl]quinolin-4-one
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-[(2-hydroxy
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-[(2,4,6-trihy one
Interactions	PUBCHEM	1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-(1H-imidaz
Interactions	PUBCHEM	1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-(1H-imidaz
Interactions	PUBCHEM	1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-3-[(3-hydroxy
Interactions	PUBCHEM	1-Cyclopropyl-3-[(2,5-dihydroxyphenyl)methyl]-7-(2,6-dimethyl-4-pyr one
Interactions	PUBCHEM	1-Cyclopropyl-3-[(2,4-dihydroxyphenyl)methyl]-7-(2,6-dimethyl-4-pyr one
Interactions	PUBCHEM	N-[3-(3,4,5-triihydroxy-benzoylamino)-phenyl]-3,4,5-trihydroxy-benza
Interactions	PUBCHEM	ethyl (2E)-2-cyano-2-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8- ylidene]acetate
Interactions	PUBCHEM	2-[1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-4-y
Interactions	PUBCHEM	2-[(2E)-2-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquin ylidene]hydrazinyl]ethanol
Interactions	PUBCHEM	3-[[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-4- dimethylpropan-1-amine
Interactions	PUBCHEM	[(E)-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-
Interactions	PUBCHEM	N-[(E)-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinoli ylidene]amino]methanamine
Interactions	PUBCHEM	(E)-1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-N-pyridin-
Interactions	PUBCHEM	(Z)-1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-N-methox
Interactions	PUBCHEM	N,1-dicyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-4
Interactions	PUBCHEM	4-[[1-Cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinolin-4-
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-phenyl-2,5- c]quinolin-3-one
Interactions	PUBCHEM	(2Z)-2-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinoli
Interactions	PUBCHEM	N-[(E)-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinoli methylmethanamine
Interactions	PUBCHEM	(E)-1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoro-N-pyridin-
Interactions	PUBCHEM	(E)-N-(4-chlorophenyl)sulfanyl-1-cyclopropyl-7-(2,6-dimethyl-4-pyrid imine
Interactions	PUBCHEM	2-Cyclohexyl-5-cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro

Main Category	Category	Terms
		c]quinolin-3-one
Interactions	PUBCHEM	N-{4-[5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-3-oxo-3 c]quinolin-2-yl]-cyclohexyl}-acetamide
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-(1-methyl- pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	[5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-3-oxo-3,5-di c]quinolin-2-yl]-acetic acid ethyl ester
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-(tetrahydro pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-(4-methyl- pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	5-Cyclopropyl-2-(3-dimethylamino-propyl)-7-(2,6-dimethyl-pyridin-4 pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-(2-hydroxy pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-[2-(2-hydro dihydro-pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-2-methyl-2,5 c]quinolin-3-one
Interactions	PUBCHEM	[5-Cyclopropyl-7-(2,6-dimethyl-pyridin-4-yl)-6,8-difluoro-3-oxo-3,5-di c]quinolin-2-yl]-acetic acid
Interactions	PUBCHEM	14-[(3S)-3-aminopyrrolidin-1-yl]-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.03,8.017,21]henicosa-2(11),3,5,7,9,13(carboxylic acid
Interactions	PUBCHEM	14-[(3S)-3-aminopyrrolidin-1-yl]-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.04,9.017,21]henicosa-2,4,6,8,10,13(21 carboxylic acid
Interactions	PUBCHEM	14-[(3R)-3-aminopyrrolidin-1-yl]-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.03,8.017,21]henicosa-2(11),3,5,7,9,13(carboxylic acid
Interactions	PUBCHEM	14-[(3R)-3-aminopyrrolidin-1-yl]-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.04,9.017,21]henicosa-2,4,6,8,10,13(21 carboxylic acid
Interactions	PUBCHEM	(5S,5aR,8aR,9R)-5-(4-amino-5-hydroxy-6-methyloxan-2-yl)oxy-9-(4-h dimethoxyphenyl)-5a,6,8a,9-tetrahydro-5H-[2]benzofuro[6,5-f][1,3]b
Interactions	PUBCHEM	Doxorubicin Hydrochloride
Interactions	PUBCHEM	(10S,13R,14R,17R)-17-[(2R,5R)-5,6-dihydroxy-6-methylheptan-2-yl]- oxo-5,6,7,8,12,15,16,17-octahydrocyclopenta[a]phenanthrene-2-car
Interactions	PUBCHEM	Topostatin
Interactions	PUBCHEM	7-chloro-2-[2-[(2-hydroxyethyl)methylamino]ethyl]anthra[1,9-cd]pyra
Interactions	PUBCHEM	(5aR,8aR,9R)-9-(5-methoxy-3,4-dioxocyclohexa-1,5-dien-1-yl)-5a,6,8 [2]benzofuro[5,6-f][1,3]benzodioxole-5,8-dione

Main Category	Category	Terms
Interactions	PUBCHEM	5-[(5S,5aS,8aR,9R)-5-(4-nitroanilino)-8-oxo-5a,6,8a,9-tetrahydro-5H-[1,3]benzodioxol-9-yl]-3-methoxycyclohexa-3,5-diene-1,2-dione
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-9-(3,4-dihydroxy-5-methoxyphenyl)-5-(4-nitroanilin[2]benzofuro[5,6-f][1,3]benzodioxol-8-one
Interactions	PUBCHEM	(5S,5aS,8aR,9R)-9-(3,4-dihydroxy-5-methoxyphenyl)-5-(4-fluoroanili[2]benzofuro[5,6-f][1,3]benzodioxol-8-one
Interactions	PUBCHEM	6,13-Bis[3-[(2-methoxyphenyl)methylamino]propyl]-6,13-diazatetracyclo[6.6.2.04,16.011,15]hexadeca-1(15),2,4(16),8,10-pe
Interactions	PUBCHEM	Di-O-(2-Thienoyl)curcumin
Interactions	PUBCHEM	1,N-diethylaminoethylaminoacetyloxy)-3-methoxyphenyl)-1,7-hepa
Interactions	PUBCHEM	1,N-diethylaminoethylaminoacetyloxy)-3-ethoxyphenyl)-1,6-heptad
Interactions	PUBCHEM	2-[4-[5-[4-(4,5-dihydro-1H-imidazol-2-yl)phenyl]furan-2-yl]phenyl]-4,5 imidazole;hydrochloride
Interactions	PUBCHEM	[4-[5-[4-(Diaminomethyl)phenyl]-2-furyl]phenyl]methanediamine
Interactions	PUBCHEM	Shikonin
Interactions	PUBCHEM	2-[4-[5-[4-(4,5,6,7-tetrahydro-1H-1,3-diazepin-2-yl)phenyl]furan-2-yl] tetrahydro-1H-1,3-diazepine;dihydrochloride
Interactions	PUBCHEM	Myricetin
Interactions	PUBCHEM	N-cyclohexyl-[1,2,4]triazolo[4,3-a]quinoxalin-4-amine
Interactions	PUBCHEM	4-Methyl-2-oxatricyclo[7.3.1.05,13]trideca-1(12),5(13),6,8-tetraene-
Interactions	PUBCHEM	12-Chloro-4-prop-2-enyl-2-oxatricyclo[7.3.1.05,13]trideca-1(12),3,5(
Interactions	PUBCHEM	4-Benzyl-4-hydroxy-2-oxatricyclo[7.3.1.05,13]trideca-1(12),5,7,9(13
Interactions	PUBCHEM	methyl 2-[8-(3,4-dihydro-2H-quinoline-1-carbonyl)-5,7-dimethoxy-2-
Interactions	PUBCHEM	VX 509
Interactions	PUBCHEM	Piroxantrone
Interactions	PUBCHEM	[2-methoxy-4-[(1E,6E)-7-[3-methoxy-4-[3-[(5-methyl-1,3,4-thiadiazol- yl)amino]propanoyloxy]phenyl]-3,5-dioxohepta-1,6-dienyl]phenyl] 3- yl)amino]propanoate
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-[bis(2-chloroethyl)amino]acetyl]oxy-3-methoxyph dienyl]-2-methoxyphenyl] 2-[bis(2-chloroethyl)amino]acetate
Interactions	PUBCHEM	Topotecan
Interactions	PUBCHEM	Nemorubicin
Interactions	PUBCHEM	Losoxantrone
Interactions	PUBCHEM	(E)-1-[(2S,3S)-3-(2,4-dihydroxybenzoyl)-4-hydroxy-2-(4-hydroxyphen yl]-3-(4-hydroxyphenyl)prop-2-en-1-one
Interactions	PUBCHEM	5-Amino-1,3,6-triphenyl-7-(pyridin-3-ylmethylimino)pyrimido[4,5-d]p
Interactions	PUBCHEM	5-Amino-7-[2-(4-chlorophenyl)ethylimino]-1,3,6-triphenylpyrimido[4,

Main Category	Category	Terms
Interactions	PUBCHEM	Picropodophyllin
Interactions	PUBCHEM	12-Bromo-4-[4-(4-pentylphenyl)triazol-1-yl]-2-oxatricyclo[7.3.1.05,1 tetraene-10,11-dione
Interactions	PUBCHEM	5-Amino-1,3,6-triphenyl-7-(2-piperidin-1-ylethylimino)pyrimido[4,5-d dithione;hydrochloride
Interactions	PUBCHEM	5-Amino-1,3,6-triphenyl-7-(pyridin-3-ylmethylimino)pyrimido[4,5-d]p dithione;hydrochloride
Interactions	PUBCHEM	5-Amino-7-[2-(4-chlorophenyl)ethylimino]-1,3,6-triphenylpyrimido[4, dithione;hydrochloride
Interactions	PUBCHEM	3-[3-(Butylamino)-2-hydroxypropoxy]-1-hydroxyxanthen-9-one
Interactions	PUBCHEM	Bis[2-[2-(6-chloro-8-oxo-14,15-diazatetracyclo[7.6.1.02,7.013,16]hexadeca-1(15),2(7),3,5,9(16),10 methylamino]ethyl] hexanedioate;dihydrochloride
Interactions	PUBCHEM	Di-O-Adamantoylcurcumin
Interactions	PUBCHEM	DI-O-Heptanoylethylcurcumin
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-(2-chloroethylamino)acetyl]oxy-3-ethoxyphenyl]-3 ethoxyphenyl] 2-(2-chloroethylamino)acetate
Interactions	PUBCHEM	N-[2-(11-methoxy-12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]he octaen-10-yl)ethyl]acetamide
Interactions	PUBCHEM	DI-O-Heptanoylcurcumin
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-[bis(2-chloroethyl)amino]acetyl]oxy-3-ethoxyphen dienyl]-2-ethoxyphenyl] 2-[bis(2-chloroethyl)amino]acetate
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-(1-adamantylamino)acetyl]oxy-3-ethoxyphenyl]-3 ethoxyphenyl] 2-(1-adamantylamino)acetate
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-[2-(dimethylamino)ethylamino]acetyl]oxy-3-ethox dienyl]-2-ethoxyphenyl] 2-[2-(dimethylamino)ethylamino]acetate
Interactions	PUBCHEM	2-[[5-Methyl-3-oxo-2-(12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca-1(17),2,4,6,8,10,13,15 enyl]amino]acetic acid
Interactions	PUBCHEM	2-[[(E)-4-methyl-3-oxo-2-(12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca-1(17),2,4,6,8,10,13,15 enyl]amino]acetaldehyde
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[3-(2-chloroethylamino)propanoyloxy]-3-methoxyph dienyl]-2-methoxyphenyl] 3-(2-chloroethylamino)propanoate
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[3-[2-(diethylamino)ethylamino]propanoyloxy]-3-eth dienyl]-2-ethoxyphenyl] 3-[2-(diethylamino)ethylamino]propanoate
Interactions	PUBCHEM	[2-ethoxy-4-[(1E,6E)-7-[3-ethoxy-4-[2-[(5-methyl-1,3,4-thiadiazol-2-yl dioxohepta-1,6-dienyl]phenyl] 2-[(5-methyl-1,3,4-thiadiazol-2-yl)ami
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-(2-chloroethylamino)acetyl]oxy-3-methoxyphenyl methoxyphenyl] 2-(2-chloroethylamino)acetate
Interactions	PUBCHEM	2-[[(E)-4-methyl-3-oxo-2-(12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca-1(17),2,4,6,8,10,13,15

Main Category	Category	Terms
		enyl]amino]acetic acid
Interactions	PUBCHEM	2-[[5-Hydroxy-5-methyl-3-oxo-2-(12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca-1(17),2,4,6,8,10,13,15 yl)hexyl]amino]acetaldehyde
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[3-(1-adamantylamino)propanoyloxy]-3-ethoxyphen ethoxyphenyl] 3-(1-adamantylamino)propanoate
Interactions	PUBCHEM	[2-methoxy-4-[(1E,6E)-7-[3-methoxy-4-[2-(4-sulfamoylanilino)acetyl dienyl]phenyl] 2-(4-sulfamoylanilino)acetate
Interactions	PUBCHEM	[4-[(1E,6E)-7-[4-[2-(1-adamantylamino)acetyl]oxy-3-methoxyphenyl] methoxyphenyl] 2-(1-adamantylamino)acetate
Interactions	PUBCHEM	2-[[5-Methyl-3-oxo-2-(12-oxo-8,14-diazatetracyclo[7.7.1.02,7.013,17]heptadeca-1(17),2,4,6,8,10,13,15 enyl]amino]acetaldehyde
Interactions	PUBCHEM	4-[(2E)-2-(2-chlorobenzylidene)hydrazinyl][1,2,4]triazolo[4,3-a]quinc
Interactions	PUBCHEM	10-(3-Aminopyrrolidin-1-yl)-11-fluoro-14-oxo-8-oxa-1-azatetracyclo[7.7.1.02,7.013,17]heptadeca-2,4,6,9(17),10,12,15-hep
Interactions	PUBCHEM	1-Cyclopropyl-3-[(2,6-dihydroxyphenyl)methyl]-7-(2,6-dimethyl-4-pyr one
Interactions	PUBCHEM	3,4,5-trihydroxy-N-[2-[(3,4,5-trihydroxybenzoyl)amino]phenyl]benzar
Interactions	PUBCHEM	N-[(E)-[1-cyclopropyl-7-(2,6-dimethyl-4-pyridinyl)-6,8-difluoroquinoli amine
Interactions	PUBCHEM	N-[4-(3,4,5-triihydroxy-benzoylamino)-phenyl]-3,4,5-trihydroxy-benza
Interactions	PUBCHEM	5-Cyclopropyl-2-(2-dimethylamino-ethyl)-7-(2,6-dimethyl-pyridin-4-y pyrazolo[4,3-c]quinolin-3-one
Interactions	PUBCHEM	14-[(3R)-3-aminopyrrolidin-1-yl]-15-fluoro-18-oxo-12-oxa-1-azapentacyclo[11.7.1.02,11.05,10.017,21]henicosa-2(11),3,5,7,9,13 carboxylic acid
Interactions	PUBCHEM	N-acetonylacety-3-(9-acridinylamino)-5-hydroxymethylaniline
Interactions	UP_KW_LIGAND	ATP-binding
Interactions	UP_KW_LIGAND	Magnesium
Interactions	UP_KW_LIGAND	Metal-binding
Interactions	UP_KW_LIGAND	Nucleotide-binding
Literature	GENERIF_SUMMARY	The heat-induced transition of TOP2A requires both Mg2+ and DNA and leads to loss of DNA strand activity. A second transition correl DNA scission, followed by religation at high temperatures.
Literature	GENERIF_SUMMARY	increased expression appears to be part of malignant cells' pheno
Literature	GENERIF_SUMMARY	topo II is not an immobile, structural component of the chromoson karyoskeleton, but rather a dynamic interaction partner of such str
Literature	GENERIF_SUMMARY	The role of topoisomerase II in the excision of DNA loop domains
Literature	GENERIF_SUMMARY	Topoisomerase II alpha represents a reliable alternative to monocl

Main Category	Category	Terms
		proliferation marker in human meningiomas.
Literature	GENERIF_SUMMARY	Overexpression of topoisomerase IIalpha induced by mutant p53 is carcinogenesis in non-small cell lung carcinoma
Literature	GENERIF_SUMMARY	Transcriptional regulation of the human topoisomerase IIalpha gen
Literature	GENERIF_SUMMARY	Redox control of cell cycle-coupled topoisomerase II alpha gene ex
Literature	GENERIF_SUMMARY	The ATP-operated clamp of human DNA topoisomerase IIalpha
Literature	GENERIF_SUMMARY	presence of dna topoisomerase II in different testicular tumor type
Literature	GENERIF_SUMMARY	Co-localizes with centromere activity and proteins within a subdon chromosome array
Literature	GENERIF_SUMMARY	Topoisomerase II-alpha expression in rectal epithelium has a signi giving topoisomerase II-targeted chemotherapeutic agents at the p reduce their mucositis side effects.
Literature	GENERIF_SUMMARY	Topo IIalpha may provide crucial information regarding selection o responsive to antineoplastic therapy, such as invasive pituitary ade carcinomas, which exhibit a high Topo IIalpha index.
Literature	GENERIF_SUMMARY	promoter elements responsible for transcriptional inhibition of pol topoisomerase IIalpha genes by p21(WAF1/CIP1/SDI1)
Literature	GENERIF_SUMMARY	A heterodimer of this protein with only one ATP binding site can go cycles
Literature	GENERIF_SUMMARY	Phosphorylation of serine 1106 in the catalytic domain of this enzy activity and drug sensitivity.
Literature	GENERIF_SUMMARY	Thr792 in human TOP2 alpha is involved in enzyme catalysis
Literature	GENERIF_SUMMARY	Analysis of TOP2A gene amplification and gene overexpression in ALL cases and inverse correlation with glucocorticoid resistance
Literature	GENERIF_SUMMARY	induction of topoisomerase IIalpha expression by peroxisome prol gamma ligands via DR1-like site is an important mechanism for th induced apoptosis
Literature	GENERIF_SUMMARY	topoisomerase II is actively exported from the nuclease and is me pathway
Literature	GENERIF_SUMMARY	Results suggest that topo IIalpha-depleted cells with the droplet-lik apoptosis, which is dependent on caspase and p53 activity during cells.
Literature	GENERIF_SUMMARY	data suggest that HER-2 amplifications are frequently linked to alte bladder cancer
Literature	GENERIF_SUMMARY	demonstrated that neither A or B isoform has any preference for a substrate under the conditions used in these experiments
Literature	GENERIF_SUMMARY	Amplification and deletion of topoisomerase IIalpha gene (TOP2A) neoplasms and may account for both sensitivity and resistance to
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha activity concentrates at heterochromatin du
Literature	GENERIF_SUMMARY	Topo-II alpha is a useful marker for diagnosing liposarcoma.
Literature	GENERIF_SUMMARY	overexpression of topoisomerase (DNA) II alpha is associated with

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Chromosomes were condensed and aligned at metaphase in topo
Literature	GENERIF_SUMMARY	alterations in the ATP binding domain of the enzyme are capable o TOP2A with DNA
Literature	GENERIF_SUMMARY	UVA-modified DNA is preferentially targeted and processed by top
Literature	GENERIF_SUMMARY	topo IIalpha destabilization depends on the newly identified domai destruction domain), which was mapped to the N-terminal 70-170
Literature	GENERIF_SUMMARY	1,4-benzoquinone stimulated DNA cleavage by topoisomerase IIal
Literature	GENERIF_SUMMARY	review commentary on TOP2A as a proliferation marker, indicator prognostic factor in mantle cell lymphoma
Literature	GENERIF_SUMMARY	Increased expression of DNA topoisomerase II alpha is associated
Literature	GENERIF_SUMMARY	topoIIalpha plays an essential catalytic role in chromosome segreg complemented by topoIIbeta
Literature	GENERIF_SUMMARY	Human topoisomerase II alpha appears to ligate the two scissile b a nonconcerted fashion.
Literature	GENERIF_SUMMARY	Cisplatin was shown to strongly inhibit the decatenation and relaxa human DNA topoisomerase IIalpha.
Literature	GENERIF_SUMMARY	low-level amplification of TIIalpha gene locus may be sporadic me protein expression in pediatric non-brainstem glioma, which can co amplification of Her-2/neu
Literature	GENERIF_SUMMARY	topo2a and HER-2 status might have therapeutic and prognostic in
Literature	GENERIF_SUMMARY	We show that a R162K mutation in human topoisomerase II alpha resistant towards vanadate while having little effect on bisdioxopip
Literature	GENERIF_SUMMARY	Further analysis of purified virus showed that HIV-1 virion containe specific kinase activities, which were partially isolated
Literature	GENERIF_SUMMARY	C-terminal domain of human topoisomerase IIalpha appears to pla modulating the DNA cleavage/ligation reaction of the enzyme and agents.
Literature	GENERIF_SUMMARY	topoisomerase IIalpha relaxes positively supercoiled plasmids >10 supercoiled molecules
Literature	GENERIF_SUMMARY	Data suggests that the alpha(2)HTH module once separated from compact conformation, integral to specific dimerization and DNA r
Literature	GENERIF_SUMMARY	Topoisomerase IIa amplifications and deletions are associated wit the adjuvant setting of breast cancer patients
Literature	GENERIF_SUMMARY	INER-37 cell line is the first cancer cell line reported with an innate region of the topo II alpha gene that confers a cytoplasmic localiza therefore an increased resistance to etoposide
Literature	GENERIF_SUMMARY	TOP2A mRNA and protein expression in ovarian cancer exhibits sp epithelial and adjacent stromal cells, which are differentially modu chemotherapy
Literature	GENERIF_SUMMARY	Increased TopoII expression after neoadjuvant anthracycline-base with relapses in breast tumors
Literature	GENERIF_SUMMARY	PKCdelta regulates topoisomerase IIalpha and thereby cell fate in

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	5/6 neuroblastoma cell lines contained 17q translocations, but the the translocated genetic material in only one of the cell lines. TOP correlate with topoisomerase IIalpha protein expression level.
Literature	GENERIF_SUMMARY	Topoisomerase II alpha and beta isoforms are present in the pre-in suggesting their significant role in HIV-1 replication.
Literature	GENERIF_SUMMARY	The purpose of this study was to evaluate amplification of topoiso as predictors of response to chemotherapy in advanced breast car amplification was present in 14 cases (16%).
Literature	GENERIF_SUMMARY	over-expression of topoisomerase IIalpha is significantly associate cervical neoplasm CIN2 to CIN3, being a late marker of cell prolifer
Literature	GENERIF_SUMMARY	topo IIalpha is a proliferation and also an apoptotic marker in atrop have a predictive value in oral lichen planus lesions prone to develo
Literature	GENERIF_SUMMARY	We conclude that topo II alpha and beta nuclear export is inhibited these proteins do not shuttle.
Literature	GENERIF_SUMMARY	During early mitosis, protein phosphatase 2A was translocated fro remained in the nucleus until pro-metaphase thus permitting the fo monoclonal 2 phosphoepitope
Literature	GENERIF_SUMMARY	HIF-1alpha is inhibited by NSC 644221 in a process that is topoiso
Literature	GENERIF_SUMMARY	Topo IIalpha and beta mRNA expression, but not the Topo IIalpha l useful marker for sensitivity to etoposide in human malignant neur
Literature	GENERIF_SUMMARY	There was a strong linear positive correlation between MCM6 labe label index in the craniopharyngiomas studied.
Literature	GENERIF_SUMMARY	40 breast carcinomas from BRCA1 mutation carriers and 40 spora examined for HER-2 and TOP2A amplification status
Literature	GENERIF_SUMMARY	Immunoreactivity for p16, MCM2, and TOP IIA was detected in only squamous epitelial lesions (LSIL-s).
Literature	GENERIF_SUMMARY	The C-terminal region of human topoisomerase II alpha determines in proliferating cells.
Literature	GENERIF_SUMMARY	The difference in DNA topoisomerase alpha labeling index betwee esthesioneuroblastoma was not statistically significant, but it was association.
Literature	GENERIF_SUMMARY	The ability of HMGB1 to interact with topoisomerase IIalpha and p binding to DNA suggests a mechanism by which HMGB1 stimulate enzyme via enhancement of DNA cleavage.
Literature	GENERIF_SUMMARY	polysomy of chromosome 17 does not affect the ERBB2 expressio mRNA expression is not related to gene status
Literature	GENERIF_SUMMARY	TOP2A status cannot be predicted from the HER-2 status and eval only in patients with HER-2 overexpression may lead to missing ca possible resistance to therapy in breast neoplasms.
Literature	GENERIF_SUMMARY	The sensitivity to depletion of topo IIalpha might be linked to struc centromere domain, indicated by shortening of the distance acros centromeres and persistence of PICH-coated connections.
Literature	GENERIF_SUMMARY	Topo IIalpha interacts with beta-catenin/T-cell factor-4 as a novel t colorectal neoplasms.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	NK314 inhibited topoisomerase II activity and stabilized topoisome complexes.
Literature	GENERIF_SUMMARY	A three-dimensional quantitative structure-activity analysis of a ne inhibitors is described.
Literature	GENERIF_SUMMARY	topoisomerase IIalpha is a novel physiological substrate for Plk3, a different roles in cell-cycle regulation.
Literature	GENERIF_SUMMARY	Data suggest that ubiquitination of topoisomerase IIalpha is deper that BRCA1 may be involved in the ubiquitination.
Literature	GENERIF_SUMMARY	Plk1-dependent phosphorylation regulates functions of DNA topoi progression
Literature	GENERIF_SUMMARY	topoisomerase IIalpha might be an effective chemotherapeutic tar carcinoma
Literature	GENERIF_SUMMARY	First detailed genome-wide characterisation of HER2/TOP2A-ampl 17q12 amplicon is complex and harbours multiple genes that may cancer development and progression, and potentially therapeutic t
Literature	GENERIF_SUMMARY	TopoIIalpha was an independent adverse predictor of failure-free s lymphoma.
Literature	GENERIF_SUMMARY	gliomas immunohistochemically express Topo-IIalpha that is corre and which is significantly less frequent in long survivors
Literature	GENERIF_SUMMARY	Analysis of TOP2A aberrations suggests a differential benefit of ac patients with primary breast cancer, favoring treatment with epirub amplifications, and perhaps deletions.
Literature	GENERIF_SUMMARY	ERCC1 was confirmed to be an independent prognostic factor for s significant role was found for ERCC1, RRM1 and TopoIIalpha in ED
Literature	GENERIF_SUMMARY	topoII-alpha could be a independent prognostic factor for a putativ
Literature	GENERIF_SUMMARY	NK314 acts as a Top2alpha-specific poison in mammalian cells, w efficacious and safe chemotherapeutic agent.
Literature	GENERIF_SUMMARY	APC has a role in the G2/M transition, potentially through associat
Literature	GENERIF_SUMMARY	Critical interaction with the 3'-bridging atom of the scissile phosph of topoisomerase IIalpha-mediated DNA cleavage.
Literature	GENERIF_SUMMARY	TOP2A deletion is associated with poor prognosis in ERBB2-ampli
Literature	GENERIF_SUMMARY	The reported poorer response to chemotherapeutic agents in patie carcinomas may stem from downregulated DNA damage response
Literature	GENERIF_SUMMARY	The topoisomerase IIalpha introduces a single-stranded break at th alpha satellite DNA in a reaction, where DNA ligation is partly unco reaction.
Literature	GENERIF_SUMMARY	Somatostatin could induce cell cycle arrest in S phase and upregul gallbladder cancer cells
Literature	GENERIF_SUMMARY	Overexpression of topoisomerase IIalpha is associated with barret high-grade dysplasia.
Literature	GENERIF_SUMMARY	expressions correlated with advance histological grading, microva age onset of the malignancy

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	findings suggest that high Topo II-alpha expression may be a usefu aggressiveness and poor outcome in laryngeal squamous cell carc
Literature	GENERIF_SUMMARY	Clusters of positively charged amino acid residues in human topoi for the enzyme to distinguish supercoil geometry during DNA relax most C-terminal cluster abrogates this recognition.
Literature	GENERIF_SUMMARY	increase in both gene expression and gene copy number in Chines carcinoma; gene amplification is associated with HER2 gene ampl
Literature	GENERIF_SUMMARY	HER2 and Topo IIalpha overexpression could be predictors of the r chemotherapy in both the CEF and CMF arms.
Literature	GENERIF_SUMMARY	These results reveal an important role of Topo II in checkpoint acti of genomic stability.
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha overexpression is significantly associated v behavior and response to drug treatment in colorectal cancer.
Literature	GENERIF_SUMMARY	Significantly higher DNA topoisomerase IIalpha is associated with leukemia.
Literature	GENERIF_SUMMARY	Data show a difference in metal ion utilization during DNA cleavag topoisomerase IIalpha and IIbeta.
Literature	GENERIF_SUMMARY	Results suggest a mechanism of up-regulation of cellular expressi HMGB1/2 in pRb-negative cells by modulation of binding of transc IIalpha promoter.
Literature	GENERIF_SUMMARY	Top IIalpha overexpression is involved in the progression of ovaria
Literature	GENERIF_SUMMARY	TEP1, TOPOIIalpha, and HSP90 interact directly with BLM in vitro a activity on telomere-like DNA substrates but not on non-telomeric s
Literature	GENERIF_SUMMARY	TOP2A gene alterations (amplifications or deletions) are associate responsiveness to anthracycline-containing chemotherapy relative that is similar to that seen in patients with HER2 amplification
Literature	GENERIF_SUMMARY	Suppression of topoisomerase IIalpha in human hTERT-RPE1 cells frequency of chromatid breaks in radiation-exposed cells.
Literature	GENERIF_SUMMARY	observations demonstrate that the QTK loop is an important playe communication in human topoisomerase IIalpha
Literature	GENERIF_SUMMARY	Topoisomerase II alpha is over expressed in 85% of renal medullar supporting the use of topoisomerase II alpha inhibitor agents to tre
Literature	GENERIF_SUMMARY	over-expressed in 4 pancreatic adenocarcinoma cases out of 13
Literature	GENERIF_SUMMARY	The majority of leiomyosarcomas had expression of topoisomeras
Literature	GENERIF_SUMMARY	In a phase II trial of a neoadjuvant platinum-containing regimen for cancer, overexpression of topo IIalpha was correlated with patholo
Literature	GENERIF_SUMMARY	TOPIIA was strongly expressed in most cases of synovial sarcoma amplification after treatment.
Literature	GENERIF_SUMMARY	These results demonstrate that Elongin B/C-cullin 2-VHL protein co ligase targeting glucose-regulated destruction domain-independen proteasome pathway.
Literature	GENERIF_SUMMARY	four acidic amino acid residues interact with metal ions in the DNA

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Studies indicate that neither HER-2 status nor topoisomerase IIalp clinically valuable in guiding prescription of anthracyclines.
Literature	GENERIF_SUMMARY	promising biomarker of metastatic breast cancer survival and pote CMF therapy.
Literature	GENERIF_SUMMARY	analysis of topoisomerase IIalpha expression in acute myeloid leu exposure to daunorubicin or ara-C
Literature	GENERIF_SUMMARY	Topoisomerase II alpha is a fundamental prognostic factor in brea
Literature	GENERIF_SUMMARY	Enhanced expression of TOP-2alpha is positively associated with a progression in Non-Muscle-Invasive Bladder Cancer.
Literature	GENERIF_SUMMARY	simple co-amplification is not the main mechanism leading to pati TOP2A aberrations and the term simultaneous amplification is mo amplification is of higher level than TOP2A amplification.
Literature	GENERIF_SUMMARY	Silencing of Bmi1 inhibits drug-induced Top2alpha degradation, inc Top2alpha-DNA cleavage complex, and increases Top2 drug effica
Literature	GENERIF_SUMMARY	A negative correlation was observed between over-expression of T metastasis in gastric adenocarcinoma.
Literature	GENERIF_SUMMARY	Increased topoisomerase II alpha is associated with prostatic carc
Literature	GENERIF_SUMMARY	Blocking the Topo II alpha gene expression by RNAi in cisplatin-res cancer cells reverses drug resistance.
Literature	GENERIF_SUMMARY	the TOP2A/CEP17 ratio may be a useful predictor of the effects of neoadjuvant chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	TOP2A gene amplification may define a subset of HER2-amplified responsible for the markedly improved chemosensitivity seen in H
Literature	GENERIF_SUMMARY	Data show that naphthalimide-based antitumor agents were prove (topo II).
Literature	GENERIF_SUMMARY	ProExC expression is not associated with HPV in squamous cell ca
Literature	GENERIF_SUMMARY	Data show that mitotic chromatin is heavily self-entangled through topoisomerase (topo) II is observed to reduce mitotic chromosom
Literature	GENERIF_SUMMARY	The expression of COX-2 was correlated with the expression of P-g significantly in gastrointestinal carcinomas.
Literature	GENERIF_SUMMARY	The expression of ToPo II, MRP and GST-pi is associated with the e adjuvant chemotherapy of gastric carcinoma.
Literature	GENERIF_SUMMARY	TOP2A gene dosage was shown to be a potent prognostic factor i
Literature	GENERIF_SUMMARY	High levels of topoisomerase IIalpha protein expression in diffuse associated with high proliferation, germinal center immunophenot treatment.
Literature	GENERIF_SUMMARY	HER-2/neu and TOPOIIalpha amplification/overexpression might d oesophagus, dysplasia and adenocarcinoma.
Literature	GENERIF_SUMMARY	TRF2, which binds preferentially to positively supercoiled DNA sub negatively regulates the amount of TOP1, TOP2alpha, and TOP2be
Literature	GENERIF_SUMMARY	cystic carcinomas of the breast have low Topo IIalpha expression EGFR protein without EGFR gene amplification.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	In vulvar squamous epithelium and its associated lesions, prolifera and Ki-67were found to be useful for vulvar squamous lesions.
Literature	GENERIF_SUMMARY	This study reports a method for the rapid over-expression and puri human topoisomerase IIalpha using the baculovirus mediated inse
Literature	GENERIF_SUMMARY	TOP2A plays an important role in the invasion and metastasis of c
Literature	GENERIF_SUMMARY	Topoisomerase II-mediatedTopoisomerase II-mediated DNA dama during the cell cycle by non-homologous end joining and homologo differently repaired during the cell cycle.
Literature	GENERIF_SUMMARY	Data show that the prognostic abilities of TOP2A and MIB-1 were s patients.
Literature	GENERIF_SUMMARY	TopIIalpha may have a role in survival for postoperative NSCLC pat chemotherapy
Literature	GENERIF_SUMMARY	Hydroquinone could repress the expression of TOPOIIalpha in hum cells.
Literature	GENERIF_SUMMARY	TOP2A coamplification, not HER2 amplification, is the clinically us incremental response to anthracycline-based chemotherapy.
Literature	GENERIF_SUMMARY	Examine inhibition of DNA topoisomerases I and II by G3 PAMAM- digoxin and proscillaridin A conjugates in a cell free system.
Literature	GENERIF_SUMMARY	Overexpression of miR-485-3p reduces expression of nuclear facto regulation of DNA topoisomerase IIalpha expression.
Literature	GENERIF_SUMMARY	In HER2-amplified breast cancer, the subgroup of patients with hor cancer and without TOP2A amplification presents the highest risk
Literature	GENERIF_SUMMARY	TOP2A expression in breast cancer has a role in high proliferation subtypes
Literature	GENERIF_SUMMARY	Topo2A has a role in chemosensitivity, and ERCC1 has a role in su tissue sarcomas
Literature	GENERIF_SUMMARY	High DNA topoisomerase II alpha in metastatic cells are associate small cell lung cancer.
Literature	GENERIF_SUMMARY	Expression and amplification of Topoisomerase-2alpha in type 1 a carcinomas are correlated with HER2/neu amplification.
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha activity and tumor cell proliferation level fro bromodeoxyuridine label in uterine body leiomyosarcoma
Literature	GENERIF_SUMMARY	TOP2A gene alterations and HER2 protein overexpression were ea responsiveness to anthracycline-containing chemotherapy in breas
Literature	GENERIF_SUMMARY	found that geminin and TopoIIalpha interact primarily in G2/M/earl that geminin recruits TopoIIalpha to chromosomal decatenation si geminin silencing in mammary epithelial cells triggers formation o suppressing TopoIIalpha access to chromosomal arms
Literature	GENERIF_SUMMARY	The expression of TOP2A correlated with higher histologic grade, t progestin receptor expression in breast carcinomas.
Literature	GENERIF_SUMMARY	Findings define a MKI67, TOP2A, and E2F1 tri-marker proliferation improved prognostication and treatment stratification in prostate c
Literature	GENERIF_SUMMARY	Expression of ER, HER2, and TOP2A displayed discordance with a

Main Category	Category	Terms
		emphasize the role of confirmatory biopsies from metastatic lesio recurrent breast cancer.
Literature	GENERIF_SUMMARY	TOP2A expression was studied in adjuvant dose-dense anthracycli Survival was longer in TOP2A-positive cases than TOP2A-negative ER/PR-negative and TOP2A-positive tumours than in other subgrou
Literature	GENERIF_SUMMARY	Results indicate that amplification of TOP2A in breast cancer is no concomitantly HER-2/neu(+), and suggest that a significant propor exhibit high levels of TOP2A.
Literature	GENERIF_SUMMARY	The expression of TOP2A was not related to the amplification of th status of the tumor. The expression of TOP2A was an independent complete remission.
Literature	GENERIF_SUMMARY	Data show that citrinin trimer tricitinol B inhibited topo IIalpha-med relaxation.
Literature	GENERIF_SUMMARY	HuR enhances TOP2A translation by competing with miR-548c-3p; control TOP2A expression levels and determine the effectiveness o
Literature	GENERIF_SUMMARY	TOP2A gene expression was significantly associated with the meta breast cancer patients.
Literature	GENERIF_SUMMARY	a new paradigm in which p38gamma actively regulates the drug-To transduction, and this may be exploited to increase the therapeutic
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha (topo IIalpha) protein expression has progn cancers. However, it is still unclear whether topo IIalpha protein ex play roles as prognostic factors in diffuse large B-cell lymphoma (D
Literature	GENERIF_SUMMARY	Meta-analysis
Literature	GENERIF_SUMMARY	Topo IIalpha expression is a valuable prognostic indicator for color
Literature	GENERIF_SUMMARY	Amplification of HER2 and deletion of TOP2A were frequently obse ploidy level.
Literature	GENERIF_SUMMARY	Conclude that EndoG and TOPO2a may actively participate in apop
Literature	GENERIF_SUMMARY	Classical lobular breast carcinoma consistently lacks topoisomera
Literature	GENERIF_SUMMARY	High TOP2A expression is associated with soft tissue sarcomas.
Literature	GENERIF_SUMMARY	This study confirms the favorable prognostic value of HER2/TOP2A
Literature	GENERIF_SUMMARY	HER-2/neu and TOPIIa are over expressed in gastric tumors and m progression.
Literature	GENERIF_SUMMARY	Loss of topoisomerase (DNA) II alpha is associated with liposarco
Literature	GENERIF_SUMMARY	The accurate cleavage of DNA by topoisomerase II is regulated thr DNA bending.
Literature	GENERIF_SUMMARY	Review of some of the published studies addressing the predictive cellular functions of this enzyme and its status in breast cancer tis
Literature	GENERIF_SUMMARY	Our findings reveal a broad involvement of Tdp1 in DNA repair and in the repair of Top2-induced DNA damage.
Literature	GENERIF_SUMMARY	Analyzed the occurrence of TOP2A gene amplification and chromo with early breast cancer.TOP2A amplification correlated with short shorter overall survival.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data suggest that microRNA/mRNA pairs in hsa-miR-140-3p/RAD5 hsa-miR-139-5p/TOP2A, and hsa-miR-133a/GCLC were correlated
Literature	GENERIF_SUMMARY	90; and under
Literature	GENERIF_SUMMARY	Amplification of Her-2/neu and/or topoisomerase IIalpha is associ in breast cancer.
Literature	GENERIF_SUMMARY	The active DNA topoisomerase II alpha is significantly less abunda deficient cells and PICH-deficient cells than at those of control cell
Literature	GENERIF_SUMMARY	Topo-IIalpha may be a novel marker of cell proliferation and/or sur cancer stem cells
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha immunoexpression in oral squamous cell c correlation with p53 expression, indicating that tumors negative to signs of p53 inactivation.
Literature	GENERIF_SUMMARY	Probing conformational changes in human DNA topoisomerase IIa mass spectrometry.
Literature	GENERIF_SUMMARY	TOP2A expression was significantly associated with recurrence in HER2 negative breast cancer.
Literature	GENERIF_SUMMARY	Msx2 and topo II-alpha may play an important role in the occurren sinonasal inverted papilloma.
Literature	GENERIF_SUMMARY	The principal site of DNA engagement undergoes highly quantized between distinct binding, cleavage, and drug-inhibited states that c subunit-subunit interactions.
Literature	GENERIF_SUMMARY	TOP2A amplification is related to anthracycline-based neoadjuvant
Literature	GENERIF_SUMMARY	A high gene dosage of TOP2A determined using qPCR occurs freq HER-2(-) tumors and has a strong adverse prognostic impact.
Literature	GENERIF_SUMMARY	Two (XRCC5 and TOP2A) of seven DNA repair and replication prot for melanoma.
Literature	GENERIF_SUMMARY	In untreated patients, Ki67, TOP2A, and RacGAP1 are significant ar markers.
Literature	GENERIF_SUMMARY	ATM-dependent TOP2A modification is required for proper regulati
Literature	GENERIF_SUMMARY	Report topoisomerase I, topoisomerase IIalpha and p53 protein ex progressive growth of ovarian tumors.
Literature	GENERIF_SUMMARY	Data indicate that STK295900 inhibited DNA relaxation activity of t and 2).
Literature	GENERIF_SUMMARY	Report topoisomerase II alpha expression in ovarian serous carcin
Literature	GENERIF_SUMMARY	show that TOP2A gene copy number alterations are not observed i protein expression of TOP2A is not related to gene amplification in
Literature	GENERIF_SUMMARY	Topo IIalpha, together with HER-2, might help to select for patients anthracycline-based neoadjuvant chemotherapy and identify non-c higher risk of disease recurrence or death.
Literature	GENERIF_SUMMARY	Mechanistic basis for chiral discrimination and explanation for the tension on eukaryotic type IIA topoisomerases.
Literature	GENERIF_SUMMARY	TOP2A is highly overexpressed in adrenocortical carcinoma and re

Main Category	Category	Terms
		invasion, and anchorage-independent growth in adrenocortical car
Literature	GENERIF_SUMMARY	studies indicate that the ability of TOP2A to prevent DNA entangle complexes and suggest that this activity contributes to the role of suppressors
Literature	GENERIF_SUMMARY	LRIG1 inhibits EGFR expression and the downstream signaling act Topo-2 expressions and eventually sensitizes glioma cells to TMZ
Literature	GENERIF_SUMMARY	MCM 7, geminin and topo IIalpha can be reliable tools for the diffe mesothelial cells and malignant mesothelioma cells.
Literature	GENERIF_SUMMARY	TOP2A overexpression significantly correlates with more advance independently predicts worse DSS and DMFS.
Literature	GENERIF_SUMMARY	These results suggest that decreases in cell proliferation due to ge inhibition of cellular topo II and that genistein, a major soy isoflavo component
Literature	GENERIF_SUMMARY	high Topo II- alpha could be identified as reliable predicators for th lesion .
Literature	GENERIF_SUMMARY	DNA topoisomerases I and II are involved in initiating APH-induce at RET
Literature	GENERIF_SUMMARY	homologous recombination seems relevant especially for Top1 an inhibitors. We also found and discuss differential pathways among inhibitors
Literature	GENERIF_SUMMARY	The mRNA expression level of STMN1, TOP2A and ERCC1 are corr carcinoma pathology type in Chinese patients.
Literature	GENERIF_SUMMARY	TOP2A gene gain was a frequent finding in colorectal cancer speci of >/=4 TOP2A copies per cell. No prognostic characteristic of TO
Literature	GENERIF_SUMMARY	The imunohistochemical staining for topoisomerase IIalpha expre additional marker for mucosal dysplasia degree determination
Literature	GENERIF_SUMMARY	Top2alpha is the major isoform mediating the replication-depende Top2alpha and Top2 mediate the transcription-dependent mechan
Literature	GENERIF_SUMMARY	The study shows TOP2B-dependent increase in the number of nuc RUNX1 and RUNX1T1 loci following etoposide treatment.
Literature	GENERIF_SUMMARY	NFBD1/MDC1 is phosphorylated by PLK1 and controls G2/M trans a TOPOIIalpha-mediated decatenation checkpoint.
Literature	GENERIF_SUMMARY	Higher TOPOIIa expression is independently associated with increa among patients undergoing surgery for clear cell renal cell carcino
Literature	GENERIF_SUMMARY	In breast neoplasms TOP2A copy number change was strongly as receptor and HER2 status and as a prognostic marker TOP2A is pr
Literature	GENERIF_SUMMARY	Results show that topoisomerase II alpha function during late G2 a for shaping mitotic chromosomes.
Literature	GENERIF_SUMMARY	The surexpression of topoisomerase II alpha was associated withi
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha, an enzyme essential for resolution of DNA binds telomeres in a TRF1-mediated manner.
Literature	GENERIF_SUMMARY	MCM2/TOP2A immunohistochemistry can help to support a diagn patients without a history of head and neck cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	TOP2 is highly expressed in proneural gliomas, rendering its pharm intratumoral administration of etoposide by convection-enhanced proneural gliomas.
Literature	GENERIF_SUMMARY	Chromosome breakage by TOP2 is an endogenous threat to gene neuronal development and maintenance.
Literature	GENERIF_SUMMARY	Based on TOP2A expression, there was no significant difference in breast cancer patients who received anthracycline and taxane-con chemotherapy.
Literature	GENERIF_SUMMARY	No TOP2A amplification occurred without HER2 co-amplification in in tissue microarrays from a clinical cohort study.
Literature	GENERIF_SUMMARY	Data suggest that TOP2A hydrolysis of neoplastic DNA in the pres by salicylate analogs; molecular models suggest this inhibition is salicylates/antineoplastic agents included in this study.
Literature	GENERIF_SUMMARY	Proteolytic degradation of Top2 enables the processing of Top2.D complexes by TDP2.
Literature	GENERIF_SUMMARY	TopoII alpha content in lung adenocarcinoma is an independent pr research of TopoII alpha with clinical and morphological paramete value in lung adenocarcinoma.
Literature	GENERIF_SUMMARY	the TOP2A gene alterations may play an important role in determin breast cancer
Literature	GENERIF_SUMMARY	TOP2A FISH and qPCR may not identify the same pathology on ch patterns may further predict outcome in breast cancer patients wit unfavorable prognosis.
Literature	GENERIF_SUMMARY	This study demonstrates that TOP2Ahigh is the phenotype of recu TOP2Aneg cells show slow cycling and have cancer stem cell prop which has significant implications for prostate cancer therapy.
Literature	GENERIF_SUMMARY	Data suggest that ability to distinguish supercoil handedness of D catalytic core of TOP2A; N-terminal gate domain of TOP2A interac inhibitors.
Literature	GENERIF_SUMMARY	Low topoisomerase II expression is associated with chemotherapy
Literature	GENERIF_SUMMARY	These results demonstrated that NPRL-Z-1 appeared to be a novel generator. Thus, NPRL-Z-1 may present a significant potential anti carcinoma.
Literature	GENERIF_SUMMARY	expression was not only associated with poorer differentiation and could also act as an independent prognostic factor in patients with carcinoma
Literature	GENERIF_SUMMARY	UHRF1 interacts with TopoIIalpha and regulates its localization to TopoIIalpha decatenates the hemimethylated DNA following replic the methylation of the nascent strand by DNMT1
Literature	GENERIF_SUMMARY	DNA damage-triggered ATF3 controlled p53 accumulation and ger breaks may serve as a switch between DNA damage and cell deat treatment.
Literature	GENERIF_SUMMARY	Combined detection of MDR-related proteins ToPo II, MRP and GS valuable for postoperative individualized chemotherapy, and furthe gastric cancer patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Activity of quinolone CP-115,955 against bacterial and human type mediated by different interactions.
Literature	GENERIF_SUMMARY	Patients screened for Top2a and Ezh2 expression would exhibit si combinational treatment involving low dose etoposide combined w
Literature	GENERIF_SUMMARY	In ACC, the DNA topoisomerase TOP2A was associated with signif disease-free survival.
Literature	GENERIF_SUMMARY	TOP2A gene amplification was not observed.
Literature	GENERIF_SUMMARY	TOPBP1 has a role in recruiting TOP2A to ultra-fine anaphase bridg
Literature	GENERIF_SUMMARY	Protein expression of BIRC5, TK1, and TOP2A in malignant periphe prognostic test after surgical resection.
Literature	GENERIF_SUMMARY	Topoisomerase-1 and -2A gene copy numbers are elevated in misr colorectal cancers.
Literature	GENERIF_SUMMARY	Mutation in TOP2A gene is associated with epithelial ovarian canc
Literature	GENERIF_SUMMARY	prognostic significance of either TOP2A expression or TOP2A gen and its various subtypes is limited.
Literature	GENERIF_SUMMARY	TOP2A expression was higher in gastric cancer from elderly patien gastrointestinal tumors from young patients.
Literature	GENERIF_SUMMARY	No TOP2A copy number changes were found in patients with de no lymphoma or relapsed DLBCL treated with chemotherapy regimens drugs.
Literature	GENERIF_SUMMARY	Report allelic imbalance at the HER2/TOP2A locus in breast cance
Literature	GENERIF_SUMMARY	JWA and topoisomerase II alpha regulate each other in tumor cells
Literature	GENERIF_SUMMARY	It might, therefore, be concluded that topoisomerase II enzyme ma radiation-induced DNA damage and consequently its inhibition con sensitizing tumour cells to ionizing radiation.
Literature	GENERIF_SUMMARY	Data show that TOP2A gene gain and protein over-expression migh platinum resistant/refractory EOC.
Literature	GENERIF_SUMMARY	b-ELE could inhibit the proliferation of HepG-2 cells and interfere w of TOPO I and TOPO IIa
Literature	GENERIF_SUMMARY	TUBB3, TOP2A, CYP19A1 and CYP2D6 gene expression, but not pr associated with patient survival in breast cancer .
Literature	GENERIF_SUMMARY	These studies revealed a relationship between TOP2A and androge that contributes to prostate cancer progression and confers sensit
Literature	GENERIF_SUMMARY	may be a useful biomarker in patients receiving adjuvant taxane-pl moderate- to high-risk endometrial cancer
Literature	GENERIF_SUMMARY	during early development, TOP2A is likely to have a role in cell prol expressed in post-mitotic cells and may be important in controlling even at this early stage.
Literature	GENERIF_SUMMARY	Positive expressions of MRP and TOP2A in the tumor tissue are as of developing brain metastases in non-small cell lung cancer (NSC
Literature	GENERIF_SUMMARY	PICH and Topo II cooperate to prevent chromosome missegregatio

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Deletion or deficiency of PTEN leads to down regulation of TOP2A, decatenation checkpoint and incomplete DNA decatenation in G2
Literature	GENERIF_SUMMARY	The study is an open label, single arm, phase II study, investigating patients with oxaliplatin refractory colorectal cancer and with a ca >/= 1.5
Literature	GENERIF_SUMMARY	we show that TopoIIalpha forms protein-protein interactions with b interacts with Wnt response elements (WREs) and promoters of di transcription, including
Literature	GENERIF_SUMMARY	TOP2A is highly expressed in advanced leiomyosarcoma (LMS)but diseases. TOP2A levels are higher in high-mitotic index tumours ar disease.
Literature	GENERIF_SUMMARY	The combined quantum and molecular mechanics calculations rev drug shows better preference in inhibiting the TOP2A compared to
Literature	GENERIF_SUMMARY	TOP2A and Ki-67 antibodies may be used in combination for cervic immunocytochemistry assays.
Literature	GENERIF_SUMMARY	Data show that comparing with Ki-67 and TOP2A, RacGAP1 allowe statement.
Literature	GENERIF_SUMMARY	This study showed that the overexpressions of Ki67, RacGAP1, and of female breast cancer patients adversely
Literature	GENERIF_SUMMARY	we propose that phosphorylation of TOP2A by CDC7/DBF4 in early localization and/or activity at centromeres, and inhibition of TOP2A prevent premature separation of centromeric DNA.
Literature	GENERIF_SUMMARY	Topoisomerase-IIalpha expression was identified as a predictor of grade papillary urothelial carcinomas.
Literature	GENERIF_SUMMARY	The methodology is useful for a high-throughput analysis of drugs just the discrimination of the Top2 isoform that is targeted but also
Literature	GENERIF_SUMMARY	RNF168 interacts with TOP2alpha to mediate its polyubiquitylation confers resistance to ICRF-193, a TOP2 catalytic inhibitor, and cyto etoposide in cultured mouse cells.
Literature	GENERIF_SUMMARY	Polyamide functionalisation at the N1-position offers a design stra properties. Dicationic HxIP* 3 increased topo IIalpha expression ar targeting agents.
Literature	GENERIF_SUMMARY	Fbxo28 regulates topoisomerase IIalpha decatenation activity and maintaining genomic stability.
Literature	GENERIF_SUMMARY	Both the genome instability and cell death of MRE11-null and MRE significantly reversed by overexpression of Tdp2, an enzyme that e conjugates; thus, the essential role of Mre11 nuclease activity is lik lesions.
Literature	GENERIF_SUMMARY	Alternative RNA Processing of Topoisomerase IIalpha in Etoposide K562 Cells
Literature	GENERIF_SUMMARY	no significant association between, MAPK and topoIIalpha express ovarian serous carcinomas
Literature	GENERIF_SUMMARY	High mRNA levels of TOP2A is independent predictor of poor outc patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	TOP2a involvement in breast cancer cells apoptosis
Literature	GENERIF_SUMMARY	HER2 amplification, but not TOP2A amplification, is a predictor of u breast cancer.
Literature	GENERIF_SUMMARY	Data indicate that cortex involvement, lower World Health Organiza topoisomerase II positivity were strong predictors for preoperative
Literature	GENERIF_SUMMARY	BD ProExtrade mark C assay containing MCM2 and TOP2A antibo nuclear staining that correlated with increased cervical dysplasia a
Literature	GENERIF_SUMMARY	This study shows that both survivin and TIIalpha are independent p human grade II/III astrocytomas stratified for IDH1-mutation status
Literature	GENERIF_SUMMARY	TOP2A rs471692 was not associated with chemoradiotherapy resp regression, weight loss, clinical stage, and cigarette smoking were predictors for these Chinese patients with non-small cell lung canc
Literature	GENERIF_SUMMARY	Tumors with higher topoisomerase IIalpha and/or mitosin express recurrence after initial treatment, and these patients may benefit fr closer radiological follow-up
Literature	GENERIF_SUMMARY	findings implicate TOP2A cleavage as a broad DNA damage mech translocations as well as a functional role of TOP2A cleavage in re elongation and gene activation.
Literature	GENERIF_SUMMARY	Study found an association between TOP2alpha gene amplificatio protein in patients with triple-negative breast cancer.
Literature	GENERIF_SUMMARY	High TOP2A expression was significantly associated with longer ti M. TOP2A and TS proteins assessed by immunohistochemistry sig mRNA expression. Immunohistochemical TOP2A expression was significant better response and longer TTP after EDP-M.
Literature	GENERIF_SUMMARY	The decatenation checkpoint is regulated, not only by topo IIalpha, also by topo IIbeta. The decatenation checkpoint is most efficient present. Deletion of most of the C-terminus of topo IIalpha, while p localization signal (NLS), enhances the decatenation checkpoint a targeted drugs. Mutation of Y640 in topo IIalpha inhibi...
Literature	GENERIF_SUMMARY	we show that despite being more cytotoxic, F14512 is less efficien TOP2alpha cleavage-complex (TOP2alphacc) in cells
Literature	GENERIF_SUMMARY	ProEx C is an immunohistochemical cocktail containing antibodies IIalpha (TOP2A) and minichromosome maintenance 2 (MCM2) pro the effective utility of ProEx C as adjunct tool in assessing the urot also providing prognostic and therapeutic information to help in cli
Literature	GENERIF_SUMMARY	These findings reveal a novel, p53-independent activity of Mdm2 a implications for the choice of chemotherapeutic agents in the trea overexpressing tumors. Herein is shown that tumor cells with MDM resistant to treatment with topoisomerase II poisons but not other
Literature	GENERIF_SUMMARY	These results explain why hTOPIIa and hTOPIIa are differentially a demonstrate the utility of C. elegans in understanding the genetics
Literature	GENERIF_SUMMARY	High TOP2A expression and Gene Amplification is associated with Carcinomas.
Literature	GENERIF_SUMMARY	HER-2 and TOP2A genes amplification are 2 separate genetic yet c cancer. Polysomy of chromosome 17 is proposed to be an early ev gene amplification.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the relation between TOP2A levels and sensitivity for doxorubicin v reports that TOP2A mRNA levels were overexpressed in MPNST ar lines exhibited relatively high TOP2A protein levels and sensitivity t
Literature	GENERIF_SUMMARY	TOP2A was identified in association with the progression and prog adenocarcinoma probably by regulating cell cycle and p53 signalin
Literature	GENERIF_SUMMARY	our data support further assessment of TOP2A and EZH2 as biom of patients with increased metastatic potential that may benefit fro targeted therapy approaches
Literature	GENERIF_SUMMARY	TOP2A acts as a co-activator of beta-catenin and activates Epithel process.
Literature	GENERIF_SUMMARY	The highly proliferating C2A subtype of hepatoblastoma is charact alpha gene up-regulation and Fanconi anemia pathway activation.
Literature	GENERIF_SUMMARY	We identified 16 potentially druggable candidates. Among them, N also found to be amplified in breast cancer, suggesting that they co the obese-deregulated transcriptome.
Literature	GENERIF_SUMMARY	Ki-67 and TOPO 2A expression correlated with tumour size and tur somatotropinomas.
Literature	GENERIF_SUMMARY	A tri-serine cluster within the topoisomerase IIalpha-interaction do required for regulating chromosome breakage in human cells.
Literature	GENERIF_SUMMARY	USP15 is required for TOP2A accumulation, and USP15 depletion l anaphase chromosome bridges.
Literature	GENERIF_SUMMARY	SENP3 phosphorylation decreased its interaction with Topo II alph deSUMOylation activity on Topo II alpha. Furthermore, we observe chromosome instability, and promotion of tumorigenesis in cells e nonphosphorylatable SENP3 mutant
Literature	GENERIF_SUMMARY	Alternative processing of TOP2alpha pre-mRNA, and subsequent s may be an important mechanism regulating the formation and/or s TOP2alpha/170-DNA covalent complexes in response to TOP2alph
Literature	GENERIF_SUMMARY	TOP2A protein showed a time dependent influence on prognosis in cancer, suggesting it might be a potential predictor of late recurren
Literature	GENERIF_SUMMARY	acetylation of specific sites involved in the allosteric regulation of mechanism for modulation of its catalytic activity.
Literature	GENERIF_SUMMARY	Data indicate that tyrosyl-DNA phosphodiesterase 2 (TDP2) alone topoisomerase II (TOP2)-DNA complexes from genomic DNA in vit in cells does not slow the removal of TOP2-DNA complexes.
Literature	GENERIF_SUMMARY	The topoisomerase II alpha expression correlated more with invas extensiveness, which might make it an eminently useful marker in pituitary adenoma behaviour.
Literature	GENERIF_SUMMARY	Steered molecular dynamics calculations suggests the Top2-catal achieved by a rocker-switch-type movement of the DNA-gate.
Literature	GENERIF_SUMMARY	TOP2A is highly expressed in liver cancer tissues and acts as an o
Literature	GENERIF_SUMMARY	We report here that both known type II topoisomerases Top2a and mammalian mitochondria, with especially Top2b regulating the su Loss of Top2b or its inhibition by ciprofloxacin results in accumula mtDNA, followed by cessation of mitochondrial transcription and r

Main Category	Category	Terms
		depletion of mtDNA copy number.
Literature	GENERIF_SUMMARY	Spearman's rank correlation analysis and Bonferroni correction we of DKI and the expression levels of GFAP, Topo IIalpha, and MGMT
Literature	GENERIF_SUMMARY	Study found both high TOP2A mRNA expression and protein expre glioma cells. High TOP2A expression in GBM was correlated with c transformation in HCMV-positive glioma cells. Also TOP2A was ide of miR-144.
Literature	GENERIF_SUMMARY	TOP2A gene alterations and CEP17 polysomy may have prognosti treated with adjuvant Anthracyclines.
Literature	GENERIF_SUMMARY	We demonstrate that Topo2a is SUMOylated in an ICRF193-depen novel non-canonical site (K1520) and that K1520 sumoylation is re segregation but not the G2 arrest.
Literature	GENERIF_SUMMARY	DNA topoisomerase II alpha (TOP2A) is associated with a longer ti following pegylated liposomal doxorubicin (PLD)-treatment. Patier treated with PLD monotherapy achieved a longer TTP compared w TOP2A status might predict activity of PLD in patients with platinu platinum-sensitive epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	nuclear division cycle 80, cyclin B2 and topoisomerase 2alpha may adrenocortical tumor development.
Literature	GENERIF_SUMMARY	Study have demonstrated that high expression rates of proliferativ in colorectal cancer is linked to a beneficial patient outcome. Acco high expression rate correlates with a poor patient outcome and st assertion.
Literature	GENERIF_SUMMARY	Doxorubicin-induced cardiotoxicity is maturation dependent due to IIalpha to IIbeta in human stem cell derived cardiomyocytes.
Literature	GENERIF_SUMMARY	Interaction network analysis of YBX1 for identification of therapeut found 12 genes playing important roles in the network formation; a FOXM1 and TOP2A were found to be in central network formation, target for therapeutics.
Literature	GENERIF_SUMMARY	Here, we combined genomic and high-throughput imaging ap-proa transcription and chromosome folding are both key contributors to instability and the formation of oncogenic MLL fusions.
Literature	GENERIF_SUMMARY	This conversion is dependent on proteasomal degradation and TD Quantitative modeling shows that only two features of pre-existing cohesin binding and transcriptional activity-can be used to predict DSBs.
Literature	GENERIF_SUMMARY	TOP2A was overexpressed in bladder urothelial carcinoma (BLCA) prognostic biomarker for BLCA. Moreover, TOP2A is functionally in invasion and survival of BLCA cells.
Literature	GENERIF_SUMMARY	The SNHG3/miR-139-5p/TOP2A axis plays an important role in the
Literature	GENERIF_SUMMARY	MiR-494, which reportedly regulates TOP2alpha, exhibited reverse combination treatment.
Literature	GENERIF_SUMMARY	Pretreatment MR-Based Radiomics Signature as Potential Imaging Expression of Topoisomerase II alpha (TOPO-IIalpha) in Rectal Car
Literature	GENERIF_SUMMARY	The human TOP2 enzymes catalyze their reactions via formation o DNA-protein crosslink, the TOP2 cleavage complex (TOP2cc). [revi

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Histone H2A phosphorylation recruits topoisomerase IIalpha to ce genomic stability.
Literature	GENERIF_SUMMARY	DNA topoisomerase IIalpha expression is decreased by miR-9-3p a resistance to antineoplastic agent, etoposide.
Literature	GENERIF_SUMMARY	Proteomic Analysis Reveals that Topoisomerase 2A is Associated Morphology.
Literature	GENERIF_SUMMARY	SPOP is essential for DNA-protein cross-link repair in prostate can
Literature	GENERIF_SUMMARY	Simultaneous Detection of Neisseria gonorrhoeae and Fluoroquino Enable Rapid Prescription of Oral Antibiotics.
Literature	GENERIF_SUMMARY	Investigation of Potential Mechanisms Associated with Non-small
Literature	GENERIF_SUMMARY	Authors demonstrated the TOP2A/UBE2C co-expression status in expression correlated with poorer prognosis.
Literature	GENERIF_SUMMARY	Inhibition of topoisomerase IIA (Top2alpha) induces telomeric DN dysfunction during chronic viral infection.
Literature	GENERIF_SUMMARY	Association of immunophenotype with expression of topoisomera acute myeloid leukemia.
Literature	GENERIF_SUMMARY	Identification of lung adenocarcinoma biomarkers based on bioinf samples.
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha prevents ultrafine anaphase bridges by two
Literature	GENERIF_SUMMARY	Topoisomerase II contributes to DNA secondary structure-mediate
Literature	GENERIF_SUMMARY	Mining topoisomerase isoforms in gastric cancer.
Literature	GENERIF_SUMMARY	CCNB2, TOP2A, and ASPM Reflect the Prognosis of Hepatocellula Weighted Gene Coexpression Network Analysis.
Literature	GENERIF_SUMMARY	Prognostic and predictive value of the combination of TOP2A and 2 cm or smaller (T1N0) breast cancer.
Literature	GENERIF_SUMMARY	FAM122A maintains DNA stability possibly through the regulation expression.
Literature	GENERIF_SUMMARY	Topoisomerase II alpha promotes gallbladder cancer proliferation activating phosphatidylinositol 3-kinase/protein kinase B/mamma signaling pathway.
Literature	GENERIF_SUMMARY	Natural Products as Anti-Cancerous Therapeutic Molecules Target
Literature	GENERIF_SUMMARY	TOP2A and CENPF are synergistic master regulators activated in c
Literature	GENERIF_SUMMARY	Implications of Topoisomerase (TOP1 and TOP2alpha) Expression Cancer.
Literature	GENERIF_SUMMARY	The ZATT-TOP2A-PICH Axis Drives Extensive Replication Fork Reve Stability.
Literature	GENERIF_SUMMARY	Neochlorogenic acid enhances the antitumor effects of pingyangm
Literature	GENERIF_SUMMARY	Expression of the Topoisomerase II Alpha (TOP2A) Gene in Lung A Association with Patient Outcomes.
Literature	GENERIF_SUMMARY	TOP2A/MCM2, p16(INK4a), and cyclin E1 expression in liquid-base

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ARID1A regulates R-loop associated DNA replication stress.
Literature	GENERIF_SUMMARY	Topoisomerase IIalpha represses transcription by enforcing promo
Literature	GENERIF_SUMMARY	Long noncoding RNADUXAP8 regulates TOP2A in the growth and r via microRNA635.
Literature	GENERIF_SUMMARY	Structural basis for allosteric regulation of Human Topoisomerase
Literature	GENERIF_SUMMARY	MiR-599 targeting TOP2A inhibits the malignancy of bladder cance
Literature	GENERIF_SUMMARY	PROGNOSTIC VALUE OF TOPOISOMERASE 2-ALPHA AND B-MYB I TREATED WITH ADJUVANT CHEMOTHERAPY.
Literature	GENERIF_SUMMARY	Expression analysis of p16 and TOP2A protein biomarkers in cervi correlation with clinico-histopathological characteristics in a referr
Literature	GENERIF_SUMMARY	Rhophilin rho GTPase binding protein 1-antisense RNA 1 (RHPN1-A carcinogenesis by sponging microRNA-485-5p and releasing DNA (TOP2A).
Literature	GENERIF_SUMMARY	Type IIA topoisomerase (TOP2A) triggers epithelial-mesenchymal t progression by regulating Snail expression.
Literature	GENERIF_SUMMARY	DNA topoisomerase II alpha promotes the metastatic characteristi transcriptionally activating beta-catenin.
Literature	GENERIF_SUMMARY	Recurrent mutations in topoisomerase IIalpha cause a previously u phenotype in human cancers.
Literature	GENERIF_SUMMARY	Genome-Protective Topoisomerase 2a-Dependent G2 Arrest Requi Cancer Cells.
Literature	GENERIF_SUMMARY	Predictive value of topoisomerase II alpha protein for clinicopathol prognosis in early breast cancer.
Literature	GENERIF_SUMMARY	Knockdown of NUSAP1 inhibits cell proliferation and invasion thro in human glioblastoma.
Literature	GENERIF_SUMMARY	Oncogenic role and potential regulatory mechanism of topoisomer analysis.
Literature	GENERIF_SUMMARY	Catalytic Mechanism of ATP Hydrolysis in the ATPase Domain of H IIalpha.
Literature	GENERIF_SUMMARY	TOP2A deficit-induced abnormal decidualization leads to recurrent NF-kappaB signaling pathway.
Literature	GENERIF_SUMMARY	E2F1-mediated up-regulation of TOP2A promotes viability, migratio apoptosis of gastric cancer cells.
Literature	GENERIF_SUMMARY	TOP2A deficiency leads to human recurrent spontaneous abortion mouse pre-implantation embryos.
Literature	GENERIF_SUMMARY	TOP2A Expression in Pheochromocytoma and Abdominal Paragan
Literature	GENERIF_SUMMARY	Topoisomerase II as a Novel Antiviral Target against Panarenaviral
Literature	GENERIF_SUMMARY	Prognostic Significance of DNA Topoisomerase II Alpha (TOP2A) i
Literature	GENERIF_SUMMARY	The clinicopathological significance of TOP2A expression in malig
Literature	GENERIF_SUMMARY	BAZ2A-RNA mediated association with TOP2A and KDM1A repres

Main Category	Category	Terms
		prostate cancer.
Literature	GENERIF_SUMMARY	O-GlcNAcylation promotes topoisomerase IIalpha catalytic activity chemoresistance.
Literature	GENERIF_SUMMARY	Targeted sequencing of cancer-related genes reveals a recurrent T DNA binding and coincides with global transcriptional changes in g
Literature	GENERIF_SUMMARY	The abscission checkpoint senses chromatin bridges through Top knots.
Literature	GENERIF_SUMMARY	cDCBLD2 mediates sorafenib resistance in hepatocellular carcinor binding to the TOP2A coding sequence.
Literature	GENERIF_SUMMARY	Role of TOP2A and CDC6 in liver cancer.
Literature	GENERIF_SUMMARY	Association of TOP2A and ADH1B with lipid levels and prognosis i adenocarcinoma and squamous cell carcinoma.
Literature	GENERIF_SUMMARY	EZH2-H3K27me3-mediated silencing of mir-139-5p inhibits cellula carcinoma by activating TOP2A.
Literature	GENERIF_SUMMARY	ERK2-topoisomerase II regulatory axis is important for gene activa
Literature	GENERIF_SUMMARY	HPV16 E6 promoting cervical cancer progression through down-re increase TOP2A expression.
Literature	GENERIF_SUMMARY	Deubiquitinase USP7 stabilizes KDM5B and promotes tumor progr resistance in nasopharyngeal carcinoma through the ZBTB16/TOP
Literature	GENERIF_SUMMARY	Activating the abscission checkpoint
Literature	GENERIF_SUMMARY	TOP2A modulates signaling via the AKT/mTOR pathway to promot proliferation.
Literature	GENERIF_SUMMARY	The correlation study between TOP2A gene expression in circulati chemotherapeutic drug resistance of patients with breast cancer.
Literature	GENERIF_SUMMARY	Expression and potential molecular mechanism of TOP2A in meta cancer.
Pathways	KEGG_PATHWAY	Platinum drug resistance
Pathways	REACTOME_PATHWAY	Transcription of E2F targets under negative control by DREAM com
Pathways	REACTOME_PATHWAY	G0 and Early G1
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	SUMOylation
Pathways	REACTOME_PATHWAY	SUMO E3 ligases SUMOylate target proteins
Pathways	REACTOME_PATHWAY	Metabolism of proteins
Pathways	REACTOME_PATHWAY	Mitotic G1 phase and G1/S transition
Pathways	REACTOME_PATHWAY	SUMOylation of DNA replication proteins
Pathways	REACTOME_PATHWAY	Post-translational protein modification
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Gastric cancer network 1
Pathways	WIKIPATHWAYS	Gastric cancer network 2
Pathways	WIKIPATHWAYS	Retinoblastoma gene in cancer
Pathways	WIKIPATHWAYS	Circadian rhythm genes
Protein Domains	INTERPRO	TopoII_euk
Protein Domains	INTERPRO	Topo_IIA
Protein Domains	INTERPRO	Topo_IIA_dom_A
Protein Domains	INTERPRO	HATPase_dom
Protein Domains	INTERPRO	TOPRIM_dom
Protein Domains	INTERPRO	DTHCT
Protein Domains	INTERPRO	Topo_IIA_bsu_dom2
Protein Domains	INTERPRO	Topo_IIA_A_a_sf
Protein Domains	INTERPRO	Topo_IIA_A/C_ab
Protein Domains	INTERPRO	Topo_IIA_B_C
Protein Domains	INTERPRO	Topo_IIA-like_dom_sf
Protein Domains	INTERPRO	Ribsml_uS5_D2-typ_fold_subgr
Protein Domains	INTERPRO	TopoIIA_CS
Protein Domains	INTERPRO	Ribosomal_Su5_D2-typ_SF
Protein Domains	INTERPRO	TOPRIM_C
Protein Domains	INTERPRO	TOPRIM_TopoII
Protein Domains	INTERPRO	HATPase_C_sf
Protein Domains	INTERPRO	DNA_Topoisomerase_II
Protein	SMART	TOP2c

Main Category	Category	Terms
Domains		
Protein Domains	SMART	TOP4c